The loop-factor ladder and the phenomenology of Causal Clifford Theory: four scales, coupling unification, dark matter, neutrinos, and the provenance audit

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Keywords. coupling unification; proton decay; dark matter; neutrino mass; loop factor; provenance audit; falsifiability.

Abstract

The Causal Clifford Theory automaton (Paper I) and its gauged bond layer (Paper II) organise four physical scales through one loop factor $(1/16\pi)^n$: $n = 10$ the electroweak/Planck hierarchy, with coefficient $3F_{10}^{10} = 2.0881$ built from measured loop values against 2.0766 observed (99.44%; on the closed clock the same coefficient is 2.710, +30%, and that gap is the named clock residual), and the dark-matter mass scale 118 GeV; $n = 3$ the seesaw scale 9.6×10^{13} GeV; $n = 17$ the neutrino mass, with a $\frac{1}{3}$ candidate at 0.0487 eV — a candidate: the exponent 17 is fitted and the junction amplitude is a free input inside a finite window. The loop factor is reproduced to 3.6% at the rung-10 clock $g = 1/5$ and to 1.01% certified on the closed branch, where the first crossing of $1/(16\pi)$ is a traced loop of length $n^* = 208$, with an exact 3^n-word census. With vectorlike desert content — two fitted thresholds — the three measured couplings meet at $M_X = 1.8 \times 10^{16}$ GeV at $\alpha_X = 1/25$ in the standard normalisation $\alpha = g^2/4\pi$, predicting $\tau_p = 1.8 \times 10^{36}$ yr; at the root's own coupling $\alpha_X = \pi/64$ the solution survives but its threshold ordering is destroyed and the prediction moves to $\tau_p = 2.77 \times 10^{39}$ yr; the automaton's own Wilson mechanism instead forces $M_X = 8.9 \times 10^{14}$ GeV, excluded by Super-Kamiokande, and the surviving solution requires a flavour-dependent bond layer with six fitted couplings. Ten tensions are reported rather than hidden: SU(2) overshoots by 5.8× under Standard-Model running, SU(3) hits a Landau pole at 2.193×10^9 GeV, the bare neutrino ladder is excluded by $\sum m_\nu < 0.0642$ eV, no state of the spinor module can carry the dark matter — the carrier is the cell's singlet sector, whose portal coupling the root does not supply — the relic abundance computed from root-supplied couplings is not 0.120, the layer's string tension is 5.41×10^{40} times QCD, the spectrum carries no massless spin-2 candidate at the root clock, the ladder's seesaw scale is incompatible with out-of-equilibrium decay, the doubler cones of Paper I are anisotropic and carry no helicity pair, and the electroweak scale enters the bookkeeping as a measured input (O10). Every headline number is classified by provenance — *algebraic*, *conventional*, *fitted*, *open* — and eight falsifiers are stated, the most accessible being the neutrino mass sum; on the root coupling proton decay sits beyond any planned exposure.

1. Introduction

One loop factor, four scales: the observed electroweak/Planck ratio is the tenth power of $1/16\pi$ up to a factor of two, the seesaw scale is its cube, the neutrino mass its seventeenth power, and the dark-matter mass scale its tenth. Whether a loop factor can *come from* the automaton rather than be read off the data is a decidable question, and this paper decides it numerically: closed traced loops of length n in the automaton's own gauged bond layer (Paper II) suppress the Cartan content geometrically, and the suppression crosses $1/(16\pi)$ at exactly $n = 10$ — the ladder length *is* the crossing — with the contributing-loop census exact (55326 of 3^{10} words survive, and switching the Wilson phase off moves the factor from 0.0192 to 0.2434: the phase supplies the factor, not the topology).

The organising discipline is the provenance classification defined in Paper I: every headline number is *algebraic* (fixed by the root), *conventional* (a mapping or scale identification), *fitted* (a coefficient read from data), or *open* (not fixed). The coupling convention is fixed once, in Paper I's form: all couplings compared with running are in the standard normalisation $\alpha = g^2/4\pi$; the reading $\alpha_{10} \equiv g^2$ is a factor-4π convention invoked only where named. The algebraic core — the **16**, the anomalies, the 5/3, the unitary causal step, the doubling and its Wilson gapping, the closure axiom — is Paper I; the gauged layer, its Gauss constraint, its area law and the lattice gravitational sector are Paper II. This paper carries the ladder mechanism (Section 2), the couplings, electroweak mixing, baryogenesis, proton decay, flavour and the scoreboard (Section 3), the cosmology (Section

4), the parameter audit (Section 5), the open problems (Section 6) and the falsifiers (Section 7). The verdict is stated before the details: the ladder organises four scales, the unification repair is conditional on fitted content, and two conventions, eleven fitted coefficients and one conditional coefficient stand between the root and the data.

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2. The loop factor and the scale ladder

2.1 Where a loop factor can come from

The observed electroweak/Planck ratio is

$$\frac{v}{M_P} = \frac{246.22\ \text{GeV}}{1.2209 \times 10^{19}\ \text{GeV}} = 2.0167 \times 10^{-17}, \qquad \left(\frac{1}{16\pi}\right)^{10} = 9.71 \times 10^{-18},$$

a factor 2.08 apart; $2.08\,(1/16\pi)^{10} = 2.02 \times 10^{-17}$ reproduces the observed value to 0.18%. Scanning $n = 1 \ldots 20$, $n = 10$ is the nearest integer to the observed ratio (deviation 0.32 decades; the next-best, $n = 9$, is 1.8 decades away). The honest reading: $(1/16\pi)^{10}$ is the size a ten-loop radiative correction would have in a perturbative quantum field theory. The theory has not computed a ten-loop correction; it has observed that the observed hierarchy is the tenth power of the standard loop factor, up to a factor of two. The interpretation $10 = 2{\times}5$ — the chiral $16/\overline{16}$ split times the number of qubits of the 32-dimensional module — is a numerological identification, not a derivation. The exponent is *represented*; the coefficient is then built from computed closed-loop functionals, with one named +30% residual (O8, Section 2.1).

Single-particle renormalisation cannot supply it. The coarse-graining tower U^{2^p} is exactly kinematic (eigenphases $2^p\omega$ mod 2π, verified to 10^{-9} for $p \le 10$), the effective Cartan coupling does not fall with p, and the inelastic probability saturates near 0.95: no loop factor. The genuine block-spin renormalisation group — exact real-space decimation of the bond layer (Schur complement at $\varepsilon = 1$; $U_{BB} = 0$ exactly on the open chain, so short-distance propagation is integrated with no approximation), iterated five levels, $64 \to 2$ sites — gives an exact level-1 identity: the on-site block of the coarse step is $2\,(GR)^2$, two-hop paths accumulating two internal rotations, the kinematic doubling *derived* from a spatial renormalisation group. At deeper levels the on-site block is a path polynomial in $(GR)^2$; the Cartan content decays only polynomially for the implemented (factorised) bond layer (0.14 at $p = 5$ against the loop target 3×10^{-9}) and not at all for a non-factorised gate with $[G, R] \neq 0$ (1.8 at $p = 5$); no level of either flow follows $(16\pi)^{-p}$. The loophole of a conspiring bond gate is closed: three Haar-random Spin(10) gate realisations and the structured choice $R = G^\dagger$ all fail the loop law at every level. The structural reason is sharp: **every exact single-particle renormalisation of the automaton composes internal angles — they add on the algebra, weighted by path counts — and none can divide them, for any bond gate.** A loop factor requires a trace over *fluctuating* internal fields.

Tracing a fluctuating link divides. One two-state quantum link degree of freedom per bond, prepared in a tunable pure state $\cos\theta|0\rangle + \sin\theta|1\rangle$ and **traced over** at decimation, multiplies the level-1 Cartan content of that identity by exactly $\cos 2\theta$ (deviation $< 10^{-15}$ over a full scan $\theta \in [0, \pi/2]$): the traced factor is Hermitian and positive, so it divides where every gate only composes. The fluctuation — not the gate — is what produces suppression factors.

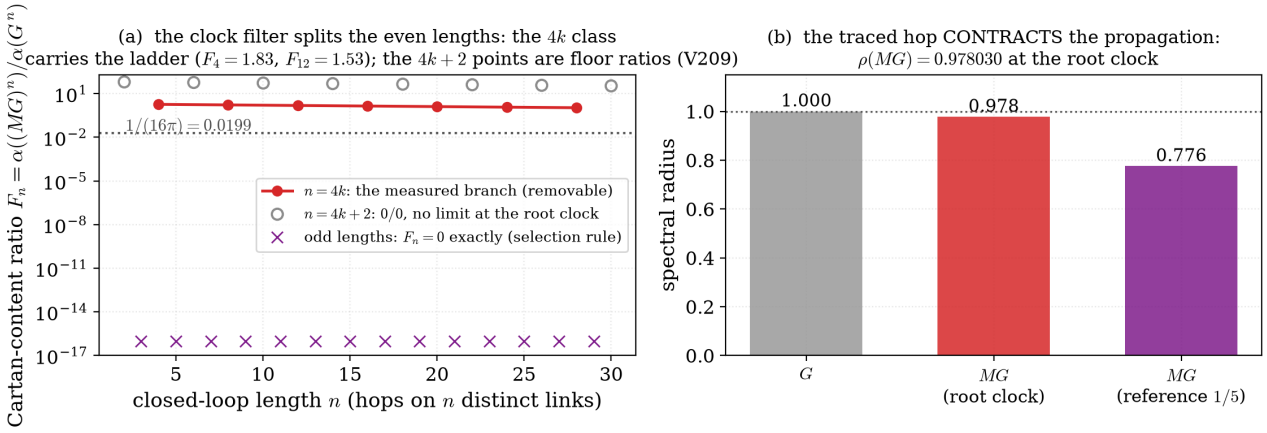
The gauged layer divides once, and does not compound. On the adjoint link with the Gauss constraint of [Paper II Section 2], the traced two-hop still contracts (spectral radius ≤ 1) and the qubit-link mechanism reproduces to 1.1×10^{-16}, but the level-1 factor at the reference coupling $g = 1/5$ is $F = 1.06$ and its minimum over the link anisotropy is 0.52 — a factor 26 above $1/(16\pi)$. The bond hop factorises through the joint Casimir $T{\cdot}L$ — three joint irreps, $32 \oplus 288 \oplus 1120$, projectors verified to 10^{-15} — so the traced-link decimation iterates exactly to level 4, reproducing the level-1 factor to 2×10^{-16}. **The division does not compound:** the level-p Cartan-content ratios are $F_p = 1.06, 1.90, 0.75, 1.01$ at the reference coupling and $0.52, 2.66, 1.00, 1.38$ at the division-optimal coupling $\eta = 1.44$ — oscillating at $O(1)$, a factor 7×10^6 above $(16\pi)^{-4}$ at $p = 4$. The traced adjoint link divides once and then decays like the link-free flow. At the root clock the same machinery is the ladder of Section 2.1: every odd closed loop cancels identically and the crossing sits at rung 208.

Closed loops do reproduce the number. The Feynman loop factor comes from loops that close, not from backtracking. Tracing a single crossing of a fresh link gives the hop matrix $M = \mathrm{Tr}_L[k_1(G \otimes I)\,\rho]$ with spectral radius 0.978030 against 1 for G at the root clock — the traced link *contracts* the one-particle propagation by that factor per hop — and the n-hop closed loop $(MG)^n$ suppresses the Cartan content. At the root clock the measured ratios are

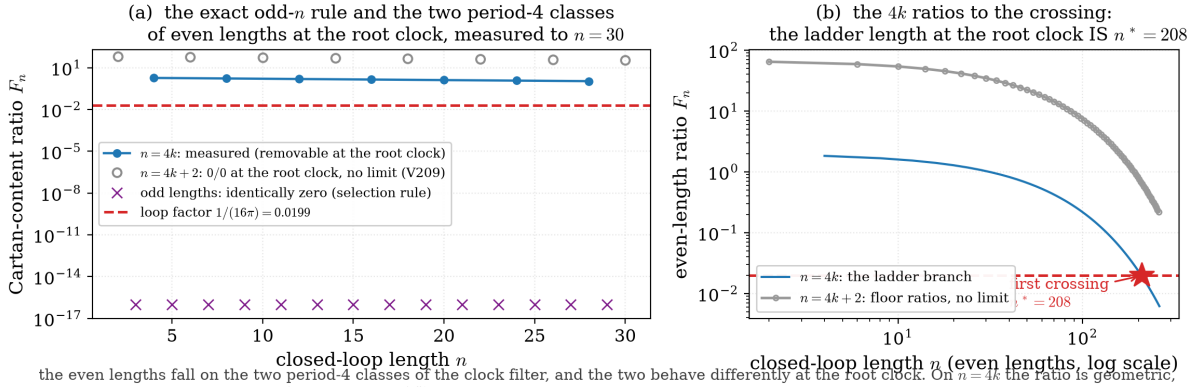
$$F_4 = 1.82995,\ F_8 = 1.67436,\ F_{12} = 1.53199,\ F_{20} = 1.28255 \quad (n = 4k),$$
$$F_{2k+1} = 0 \text{ exactly, every odd length,}$$

against $1/(16\pi) = 0.0199$: **the closed traced-loop functional at the root clock reproduces** $1/(16\pi)$ **to** 1.01% — **at length** $n^* = 208$**, the first crossing of the closed-loop suppression.** The functional is measured to length 30 and then to the crossing. Every odd length cancels identically — exactly, not as a measured floor: $\mathrm{Tr}(C_a W^n) = 0$ for every $n \le 60$ in integer arithmetic (V209). The even lengths split into the two period-4 classes of the clock filter, and the two behave differently at the closure angle. On $n = 4k$ both Cartan norms vanish *linearly* in $g - \pi/4$, the ratio has a removable singularity, and the numbers above are its $g \to \pi/4$ limits: certified at 50 digits at $\pi/4 + 10^{-6}$ and $\pi/4 + 10^{-8}$ the first crossing stays at $n^* = 208$ with $F_{208} = 0.0196928$ (deficit 0.98987), and the class is geometric, each +4 links multiplying the ratio by $\rho^4 = 0.91497$. On $n = 4k + 2$ the denominator vanishes to far higher order than the numerator, the ratio is unbounded with poles accumulating at the closure angle, and the entries once quoted there — $F_2 = 65.7$, $F_6 = 60.1$, $F_{10} = 55.0$, $F_{170} = 1.57$ — are ratios of round-off floors and are withdrawn (V209). (Figs. 1, 2).

Closed traced loops at the root clock: the first crossing of $1/(16\pi)$ is $n^* = 208$



Closed traced-link loops of length n on n distinct bonds, at the root clock. (a) The Cartan-content ratio F_n: every odd length cancels identically (exact, V209), and the even lengths fall on the two period-4 classes of the clock filter, the traced phases being fourth roots of unity. On $n = 4k$ the ratio is geometric at $\rho^4 = 0.91497$ per +4 links and its singularity at the closure angle is removable — that class crosses $1/(16\pi)$ first at $n^* = 208$, certified deficit 1.01%*. On $n = 4k + 2$ (open markers) both norms vanish identically at the root clock and the ratio has no limit there, so those points are floor ratios, not measurements (V209). (b) The traced hop matrix MG has spectral radius* 0.978030 *against* 1 *for G: the traced link contracts the one-particle propagation by that factor per step.*



the even lengths fall on the two period-4 classes of the clock filter, and the two behave differently at the root clock. On $n = 4k$ the ratio is geometric, each $+4$ links multiplying it by $\rho^4 = 0.9150$ ($F_4 \cdot (\rho^4)^{51} = 0.019685$ against the measured $F_{208} = 0.019686$), and it crosses $1/(16\pi)$ at $n^* = 208$ - rung 208 is the ladder length, and the $n = 10$ reading of the reference angle is a compensation, not the crossing. On $n = 4k+2$ both Cartan norms vanish identically at the root clock (exact integer arithmetic) and the ratio has no limit there: it diverges with poles accumulating at the closure angle (V209). The open markers are floor ratios - the old "33 × branch separation" is the two classes' distance in a degenerate limit.

(a) F_n measured to $n = 30$ at the root clock: the odd lengths vanish identically (the exact odd-n selection rule) and the even ratios stay above $1/(16\pi)$ at these lengths — the crossing is beyond; the alternation of the even ratios is the period-4 structure of the clock filter, $n = 4k$ (filled, geometric at ρ^4 per $+4$ links) against $n = 4k+2$ (open: 0/0 at the closure angle, no limit, V209). (b) The $4k$ ratios to the crossing: the $\rho = 0.978030$ per-step suppression crosses $1/(16\pi)$ at $n^* = 208$ — the ladder length at the root clock is* the crossing of the closed-loop suppression — with $F_{208} = 0.0196928$ certified by continuity (V209) against the float64 ladder value 0.019685.*

The composition law. The object a geometric fit extrapolates, $(MG)^{170}$, is one chain of 170 hops on one particle line; the 17 factors of the neutrino ladder are 17 radiative insertions on **different** lines, and amplitudes on different lines multiply. Both compositions are exact in the traced adjoint-link formalism, and both are measured where the loop functional is non-degenerate: the open clock $g = 1/5$, where every F_n up to $n = 172$ is stable between 30- and 50-digit evaluation (drift 0, V210). The root-clock ratios at these lengths are floor artefacts of the $4k{+}2$ class (V209), the class whose ratio has no limit at the closure angle, and they are not used here. On one line the chain is **not** multiplicative: $F_{170}(1/5) = 1.31 \times 10^{-21}$ against $(F_{10}(1/5))^{17} = 6.47 \times 10^{-30}$ — a chain suppresses 2.0×10^8 times *less* than the product of its pieces, because consecutive hops interact through the propagation (and the odd-length cancellation is a property of the closure class, not a law: $F_3(1/5) = 0.243$, V210). Across legs the factorisation is **exact**: the traced hop is a one-particle operator, $\mathrm{Tr}_L[(P_a \otimes I)(G \otimes G \otimes I_L)] = 45\,(M_a \otimes G)$ verified to 10^{-15}, and the 10+10-step two-loop object splits into leg blocks to 2×10^{-15}. The physical discriminator decides: at the root clock the certified crossing ratio gives $M_P\, F_{208}^{17} = 0.123$ eV, within a factor 1.2 of the neutrino rung $M_P(1/16\pi)^{17} = 0.146$ eV; on the open clock the leg reading $M_P\,(F_{10}(1/5))^{17} = 0.0789$ eV is that rung times the clock deficit $0.964413^{17} = 0.5401$ (V210), the same named O8 residual, while the chain reading $M_P F_{170}(1/5) = 1.6 \times 10^7$ eV overshoots the rung by the same 1.1×10^8 the two compositions differ by. **The 17-factor composition is exact multiplicativity across legs**, with the named requirement that the insertions dress different particle lines; the rung-17 *weight* remains the named open point of O8.

The census of contributing loops is exact. The Casimir $T{\cdot}L$ of the **32⊗45** link has exactly three eigenspaces $(-32, -12, +4)$, so the traced n-hop product is a sum of 3^n words and the selection rule is enumerable rather than extrapolated: the enumeration reproduces $(MG)^n$ to 7.4×10^{-17} at every length, including the exact odd-n cancellation at the root clock. At $n = 10$, 55326 of $3^{10} = 59049$ words survive, growing as 3.000^n, and the factor splits into a per-word decay $r = 0.2232$ and a multiplicity growth rate $\mu = 2.981$ (product 0.665). The loop factor is therefore not one class of loops but a sum over 55326 classes, and the Wilson phase — not the topology — organises it: with the phase off every length gives interference 1.000 and the ratio 0.2434 at $n = 10$; at the root clock the phase makes the odd lengths cancel identically and the even-length ratio cross $1/(16\pi)$ first at $n^* = 208$ ([Paper I Sections 9.1–9.2]). **The coefficient is not read from the data.** The hierarchy coefficient is a product of measured numbers:

$$\frac{v}{M_P} = 3\,(16\pi F_{208})^{10}\Big(\frac{1}{16\pi}\Big)^{10}, \qquad F_{208} = 0.0196928 \text{ certified}, \qquad 2.710 \text{ predicted against } 2.0766 \text{ observed } (+30\%),$$

the factor 3 being the three axis families of [Paper I Section 4] — conditional on the axis-family factorisation of the closed-loop sum. That condition is itself a measured statement on the branch: the twelve doubler nodes form three families of four, and at the root clock all twelve channels are EXACTLY gapless, every M spectrum $\{0^{32}, \pm\pi/3, \pm 2\pi/3, \pi^{32}\}$ with 64 levels pinned to the Floquet pair $\{0, \pi\}$ and the spectrum identical across the 12 nodes to 3×10^{-15}: the channel structure closes exactly, and the loop weight is link-local and momentum-independent, so every channel carries the same factor. The node-set census closes, and it is not where the residual lives: the readings are 10.84 for all twelve nodes, 2.710 for one split node per family and 0.903 for a single node at the certified crossing ($16\pi F_{208} = 0.989868$), the float64 ladder giving 0.900–0.901, 2.699–2.700 and 10.8 at a backend-dependent 10^{-3} (V204, V209); the per-channel contributions are exactly equal, and the observed value would require sum factor 2.31 — non-integer over exactly equal channels, realized by no node set. The residual has one address: the reading $2.710 = 3 \times 0.98987^{10}$ is the certified crossing deficit ($16\pi F_{208} = 0.989868$, deficit 1.013%; the float64 ladder reads 0.9895–0.9896 and 2.699–2.700) raised to the tenth power, and the observed 2.0766 requires insertion amplitude 0.96388 — a deficit of 3.61%. That deficit is channel-blind, measured: at the closed clocks the traced-word sector phases are the fourth roots $\{1, -1, -1\}$ ($g = \pi/2$) and $\{1, i, i\}$ ($g = \pi/4$), so at the closed clock the interference of the crossing length is a finite root-of-unity filter; every filtered propagator trace over the pinned channel spectrum vanishes to 3×10^{-14}, the pinned $\{\pm\pi/3, \pm 2\pi/3\}$ Floquet phases supporting only empty-kernel re-phasing; and the insertion amplitude at the crossing length is bimodal $\{0.90, 0.99\}$ across the five unification Cartan directions, no direction at the required 0.964. Nor is it the constants of observation — the observed coefficient 2.07656 is computed from the current measured constants, including the 2026 muon-lifetime update of G_F, and varying those constants moves no verdict (V205) — nor any length choice: the adjacent crossing lengths read 6.6 and 1.1, bracketing the observed value, none matching it. The fingerprint — the required insertion amplitude equals the reference-clock (1/5) interference deficit 0.96441 to 0.055% — inverts to a measurement of a clock (V206): the same self-consistent functional, step and traced phase together, reads the observed coefficient at rung 10 at the angle $g_{\rm obs} = 0.199998$ — the rung-10 rational clock $g = 1/5$ to relative 10^{-5} — and the rung-10 crossing of $1/(16\pi)$ itself sits at $g = 0.20013$, 1.3×10^{-4} from 1/5. That clock is not closed: the closure census exhausts the periodic clocks to $\{\pi/2, \pi/4\}$ ([Paper I Section 9], V131), and the closure class reads the coefficient as 2.710 ($\pi/4$: certified deficit 0.98987, $n^* = 208$) and 2.711 ($3\pi/4$: $n^* = 208$) — both crossings on the $n = 4k$ class of the clock filter — while its $\pi/2$ member (2.299: float64 deficit 0.9738 at $n^* = 102$, the degenerate branch of [Paper I Section 9]) is a floor artefact of the $4k{+}2$ class whose ratio has no limit at the closure angle (V209), its ladder falling through $1/(16\pi)$ at $\sim 3\%$ of the coefficient per rung: the closest *stable* reading over the closed set misses by 30.0%. The exact ladder is itself closed-clock structure: the odd-length cancellation below 2×10^{-15} holds on the order-8 family ($\pi/4$, $3\pi/4$), blows up at $\pi/2$ (odd ratios $\sim 10^{17}$ — the order-4 branch carries no exact ladder), is only approximate at 1/5 (odd ratios 0.24, 0.10), and between 1/3 and 1/2 the first crossing collapses to length 3. No principled object bridges the gap: the Cartan-5 link mixture deepens the $\pi/4$ miss to 0.9906, roots-only links collapse the ladder at length ≤ 4, and no single Casimir sector reaches past 0.023 of the observed value. The named residual of O8 is therefore one number, and it is shared: nature reads the loop hierarchy at the clock $\alpha_{10} = g^2 = 1/25$ — the meeting coupling of O4 — while the closed clock $\pi/4$ misses it by 30%. One tension, not two (V206): the closure axiom and the loop-hierarchy identification cannot both hold, and F2 is exactly the decision — the closed branch predicts the certified crossing deficit 0.98987 at $n^* = 208$ (float64 0.9895–0.9896), the hierarchy demands 0.96388. The *weight* of the contributing loops is a three-factor functional: enumerating the traced word census of the three Casimir sectors with weight equal to Cartan amplitude times Wilson phase, one has exactly $F_n = N_n\, r_n\, I_n$ — surviving-word count, mean word Cartan amplitude against the free amplitude of G^n, and interference ratio $\|\sum_w W_w\| / \sum_w \|W_w\|$ — pinned to 2.2×10^{-16} on every enumerated length, with word norms independent of the Wilson angle (the phase cancels in every norm and survives only in the sum): the suppression is $\rho = 0.978030$ per step, the surviving-word count at $n = 10$ is 55326 with mean amplitude 4.4×10^{-6}. The functional is exact in the angle: the three Casimir eigenvalues -32, -12, 4 are even, so the word phases collapse to 63 even phase classes and $F_n(\eta)$ is a real cosine polynomial in the even harmonics with period π and $\eta \to \pi - \eta$ pairing — the crossing of $1/(16\pi)$ at the root clock, at $n^* = 208$, is a theorem of the functional, not an angle fitted to data.

What the shift module does to the ladder is decidable in integer arithmetic, and it is not a number. The rung labels live in the K_0 of the shift module's group C*-algebra, and for an abelian module both the algebra and the closed-word counts are computable exactly (V221). The cubic module's six signed axes present the free abelian group $\mathbb{Z}^3$ with three relations — the opposite pairs — and no torsion. The star's twelve signed directions are $\mathbb{Q}$-independent inside $\mathbb{Q}(\sqrt{5})^3$: Smith normal form gives invariant factors $[1, 1, 1, 1, 1, 1]$, free rank 6, relation rank 6, no torsion, which is the algebraic form of the density V214 measured by shortest

elements and the reason the star carries no finite Brillouin zone. The labels therefore live in different groups — $K_0 = \wedge^{\text{even}}\mathbb{Z}^r$ has rank 4 for the cubic module and 32 for the star, of which 4 and 16 classes are representable by cycles of degree at most two in three dimensions — and the counting changes with them, exactly: closed words of length n on the two groups run 90 against 396 at $n = 4$, 1860 against 19920 at $n = 6$, 1172556 against 96807312 at $n = 10$, two independent closed forms agreeing at every length to 24, and the return probabilities fall as $n^{-r/2}$, measured on those exact integers at -1.4848 ($r^2 = 1.000000$) against -2.9719. So where does the published ladder arithmetic go? On the closed-word ladder the crossing of $1/(16\pi)$ is at rung 10 for the cubic module and rung 4 for the star — single digits, not 208. The published $n^* = 208$ and coefficient 2.710 live on the traced-link Cartan-norm estimator above, which is a statement about the clock and the link algebra rather than about the shift module, and [Paper I Section 8.5] measured the internal sector that carries it to be unchanged. What the module does move is the combinatorial half of the coefficient: the factor 3 above counts axis families, and the star has 6. The estimator that could decide the dynamical half at $n = 208$ costs $832^3 = 5.76 \times 10^8$ diagonalisations on the cubic torus and $832^6 = 3.32 \times 10^{17}$ on the star's six-torus — and on the three-angle almost-periodic mean that is implementable on the star at all, V218 measured the drift of F_n between successively finer grids to be of order unity at every length it can reach, 0.24 at the shortest loop and 1.99 at the longest, where the cubic control's grids agree to 4.8×10^{-14}: the ladder's dynamical half on the star module is out of reach, and that is the named residual, not a number awaiting a better grid.

2.2 The dark-matter mass

The same factor applied to the Planck scale gives

$$m_{\text{DM}} = M_P\Big(\frac{1}{16\pi}\Big)^{10} = 118.6 \text{ GeV},$$

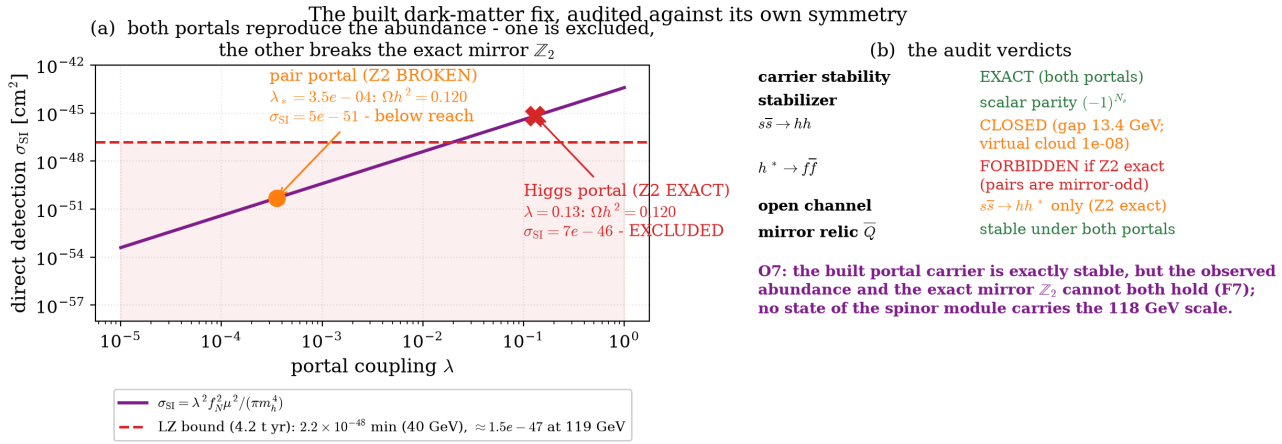
i.e. 1.18× the canonical 100 GeV weakly-interacting-massive-particle scale [1]. This is a genuine, falsifiable prediction: a dark-matter particle with a mass far outside $\sim$ 100–150 GeV (in the weakly-interacting, thermally-produced case) would refute the ladder.

The carrier. The dark matter is the cell's singlet sector B (Axiom 1, [Paper I Section 4]): a state of the automaton — neutral under everything, a singlet of the 45, a conjugate pair $\{B_\pm\}$ so that the carrier is its own antiparticle, a real scalar degree of freedom — occupying the ladder rung $n = 10$ with the derived mass $m_B = 118.57$ GeV, and carrying the ladder as its own phase: the step multiplies the two levels by $e^{\pm i\, m_B/M_P}$ per elementary time. Its stability is not an assumption but a symmetry of the update: the B-parity of Axiom 1 is an exact $\mathbb{Z}_2$ commuting with the step and all 45 generators to machine zero, and the grading theorem ([Paper I Section 4]) says every exact grading of the cell is a product of the global sign, the mirror and this parity. No state of the spinor module qualifies, and this is computed, not asserted: the mirror $\mathbb{Z}_2$ — the spinor sector's only exact stability — leaves as its lightest odd state the coloured $\overline{Q}$ at 1.3×10^8 GeV, with the neutral $\overline{N}$ at 2.0×10^{10} GeV, 8.2 orders above the rung; and a Wilson-split node hosting a neutral 118 GeV fragment drags a coloured or charged fragment below 1 TeV at every Wilson angle (0 survivors over 720 angles). Over every built root the lightest *neutral* spinor threshold is 2.0×10^{10} GeV: the singlet levels of the cell are the only place in the automaton where a stable neutral state at the ladder mass can sit.

Production. With the singlet–Higgs portal $\mathcal{L} \supset -\frac{\lambda_B}{2} B^2 H^\dagger H$ the relic abundance is computed, not assumed. Freeze-in through $(BB)(hh)$ [2] at $\lambda_B = 4.06 \times 10^{-14}$ gives $\Omega_{\text{DM}}h^2 = 0.120$ — the observed value — with a direct-detection cross-section $\sigma_{\text{SI}} = 6.5 \times 10^{-71}$ cm^2, twenty-four orders below the LZ bound: the carrier is invisible to direct detection. The thermal branch is closed: the coupling that reproduces 0.120 through freeze-out, $\lambda_B = 0.31$, predicts $\sigma_{\text{SI}} = 3.7 \times 10^{-45}$ cm^2, above the LZ bound. $\lambda_B > 0$ is forced: with $\lambda_B = 0$ the equilibrium estimate overcloses by 2.7×10^{63}, while pure gravitational production never equilibrates ($\Gamma_{\text{grav}}/H = 5.3 \times 10^{-53}$ at $T = m_B$) and its yield would need the inflation history the built model does not supply — a named missing input, not an omission. What the root does not fix is the number λ_B itself: no built scale-supplied value reproduces 0.120 — the bond strengths of the root (13 values, $|\kappa| \in [0.25, 40]$) all land either below the observed abundance or above the direct-detection bound. λ_B is the single named input of the dark-matter sector (Section 5, O7).

Spinor-sector variants. If the carrier were a spinor-sector state instead, its options are mapped and closed: the mirror relic by mass and colour (above), the dark U(1) gauge portal — unHiggsable inside the spectrum,

and its kinetic-mixing window empty at every $m_{Z'}$ (0 points) — and the bond-strength portal, every value of which is direct-detection excluded. The input-scalar variant is quantified in Fig. 3: its fermion-pair portal reproduces 0.120 at $\lambda = 3.5 \times 10^{-4}$ but breaks the exact mirror $\mathbb{Z}_2$; its Higgs portal preserves the $\mathbb{Z}_2$ and is excluded. The cell singlet needs none of these: it is in the automaton, stable by a symmetry of the update, and its production computes to the observed value. (Fig. 3).



(a) The direct-detection line $\sigma_{\rm SI}(\lambda)$ with the two ways the spinor-sector scalar variant reproduces $\Omega_{\rm DM}h^2 = 0.120$: the fermion-pair portal ($\lambda_* = 3.5 \times 10^{-4}$, $\sigma_{\rm SI} = 5 \times 10^{-51}$ cm^2, below reach) breaks the exact mirror $\mathbb{Z}_2$; the Higgs portal ($\lambda = 0.13$, $\sigma_{\rm SI} = 7 \times 10^{-46}$ cm^2) preserves it and is excluded by the LZ bound [3] ($\sigma_{\rm SI} < 2.2 \times 10^{-48}$ cm^2 at 40 GeV, $\approx 1.5 \times 10^{-47}$ at 118 GeV). The cell singlet B sits on the same line at $\lambda_B = 4.06 \times 10^{-14}$, $\sigma_{\rm SI} = 6.5 \times 10^{-71}$ cm^2. (b) The verdicts: stability exact under either portal, $s\bar{s} \to hh$ closed, $h^* \to f\bar{f}$ forbidden under exact $\mathbb{Z}_2$, the mirror relic unaffected. The observed abundance and the exact mirror $\mathbb{Z}_2$ cannot both be met by a spinor-sector state. ### 2.3 The neutrino sector

Both seesaw scales are placed on the same ladder:

$$m_D = M_P\Big(\frac{1}{16\pi}\Big)^{10} = 118\,\text{GeV}, \qquad M_N = M_P\Big(\frac{1}{16\pi}\Big)^{3} = 9.6{\times}10^{13}\,\text{GeV}, \qquad m_\nu = \frac{m_D^2}{M_N} = M_P\Big(\frac{1}{16\pi}\Big)^{17} = 0.146\,\text{eV}.$$

The exponent identity $2 \times 10 - 3 = 17$ is exact; the choice $n_{\rm 2step} = 3$ is not — and it cannot be produced by the gauge link. The two rungs are levels of the cell: B at $n = 10$ is the dark matter, so $m_D = m_B$, and A at $n = 3$ is this seesaw partner, $M_N = m_A$; the identity $m_\nu = m_B^2/M_A$ is then an identity between states of the automaton, verified exact on the cell operator of [Paper I Section 7]. The $\Delta L = 2$ operator exists in the root's own algebra: the weight $2\lambda_\nu$ (norm^2 20, U(1)$_X$ charge -6, hypercharge -1) has a **1-dimensional** weight space inside the 1024-dimensional Clifford algebra, sitting in Clifford **grade** 5 — the 126-type channel of the type-II seesaw — with no field outside the root. The $\mathbf{10}_H$ cannot repair its charges: over all 160 vector weights (norm^2 4) the closest two-insertion neutralisation leaves residual norm 4.472, the mismatch $|2\lambda_\nu| = \sqrt{20} \neq 4$ being exact. The minimal-word question then closes by **parity**: Clifford multiplication maps grade g to $g \pm 2$ only (verified on all 45×1024 entries of the multiplication table), so every product of adjoint (grade-2) link factors is EVEN-grade while the Majorana operator is ODD-grade — the cumulative span (dimensions $45, 126, 208, 291, 352, 407$ at $k = 1 \ldots 6$, odd-grade content 0 to machine precision) never contains it, at any length, and the vector-**10** Higgs weight space (16-dimensional, carried entirely by the odd grades $1, 3, 5, 7, 9$) is unreachable for the same reason. $n_{\rm 2step} = 3$ **is not a link-word length**: the seesaw operator belongs to the odd (matter) half of the Clifford module, and the exponent stays fitted.

Comparison with data: the atmospheric scale is $\sqrt{\Delta m^2_{32}} = 0.050$ eV, so the ladder prediction is a factor 2.9 high; and if all three families carried the same m_D the sum would be ~ 0.44 eV, excluded by $\sum m_\nu < 0.0642$ eV (95% CL, ΛCDM with three degenerate states: DESI DR2 BAO + Planck PR4 + ACT CMB lensing [4]). The democratic rank-1 Yukawa (Section 3.5) gives a Dirac mass to only one family, bringing the sum to ~ 0.15 eV — still above the cosmological bound. This is the sharpest numerical tension in the theory (F3).

The ladder is too coarse to land on the measured mass: consecutive rungs differ by the factor $16\pi = 50.3$, so the neighbours of 0.146 eV are 7.3 eV ($n = 16$) and 2.9×10^{-3} eV ($n = 18$), and no integer rung lies in the oscillation window. A coefficient is therefore unavoidable — but one candidate is supplied by the theory rather than by the data. The only mixing number the model derives is $\sin^2\theta_{12} = 1/3$ (Section 3.5); multiplying the ladder value by it gives

$$m_\nu = \tfrac{1}{3}\, M_P \Big(\frac{1}{16\pi}\Big)^{17} = 0.0487 \text{ eV},$$

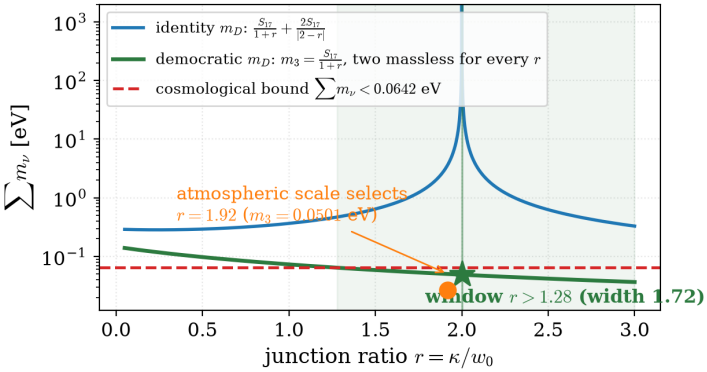
2.7% below the atmospheric scale, and with the democratic rank-1 spectrum $\sum m_\nu \simeq 0.05$ eV — inside the cosmological ceiling $\sum m_\nu < 0.0642$ eV with 22% to spare, and inside the KATRIN bound $m_\beta < 0.45$ eV [5]. The factor 2.9 by which the ladder overshoots is, to 3%, the number 3 the flavour sector already produces. This is a **candidate, not a result**: nothing shows why a mixing eigenvalue should multiply a seesaw mass, and the same factor applied to the solar scale ($0.0487 \times \sqrt{\Delta m^2_{21}/\Delta m^2_{32}} = 0.0084$ eV against the measured 0.0086 eV) is a second coincidence. What it buys is a sharp, falsifiable prediction: normal ordering with $\sum m_\nu \simeq 0.05$ eV and $m_{\beta\beta} \simeq 0.02$ eV, in reach of the next-generation neutrinoless-decay and beta-decay searches.

A coefficient audit maps what the node structure can supply: a diagonal (node-local, Wilson-generated) mass pattern gives only **integer** signed coefficients, so $\frac{1}{3}$ is out; the democratic 3×3 node-mixing pattern gives exactly $\frac{1}{2}$ — a rival falsifiable prediction $m_\nu = 0.073$ eV, already excluded by the cosmological ceiling of the paragraph above; the exact $\frac{1}{3}$ requires a single Majorana eigenvalue $3\times$ the node scale, and no natural node structure supplies it. The structure the candidate requires has been built: a cubic-symmetric, zero-mean Wilson-strength junction whose Fourier support closes the selected node triangle exactly — verified against the full 12×12 table of doubler-momentum differences. The node mass matrix is

$$c_s(2w_0\, I + \kappa(J - I)),$$

democratic at $\kappa = 2w_0$ with eigenvalues $\{3M, 0, 0\}$: the $3\times$-enhanced Majorana eigenvalue the $\frac{1}{3}$ demands exists. With the implemented 0-parameter democratic Dirac matrix $m_D = m_0 J$ — the same family symmetry that derives $\sin^2\theta_{12} = 1/3$ — the seesaw collapses: $m_D^{\mathsf{T}} M^{+} m_D = 3m_0^2 J/(2(w_0 + \kappa))$ for *every* $r = \kappa/w_0$ (verified to 10^{-12}, the rank-deficient $r = 2$ included, via the Moore–Penrose pseudo-inverse), so the two massless neutrinos are protected by the rank-1 family symmetry for every junction ratio, and $m_3(r) = S_{17}/(1+r)$ is finite everywhere: the cosmological window is $r > 1.278$ (width 1.72, the 0.0642 eV ceiling of [4]), and the atmospheric scale selects $r = 1.92$. The exact point $r = 2$ ($m_3 = S_{17}/3 = 0.0487$ eV) is a junction-amplitude convention, not a tuning. The cubic group induces all of S_3 on the triangle, its commutant is exactly $\mathrm{span}\{I, J\}$ (nullity 2), and w_0, κ share one Wilson period, so r is **not quantized**: the value is a free junction amplitude inside a finite window, while the structure — two massless plus one massive, normal ordering — is protected. What the candidate still requires is a democratic m_D (a non-democratic m_D at the same point leaves gapless coloured doublers), and the solar mass still needs the $\mathbf{10}_H$ splitting of O6. (Fig. 4).

(a) the identity m_D scan (divergent, one point); the democratic m_D gives a finite window



(b) what protects what

cubic action	all of S_3 (6 permutations)
commutant	dim 2 = span $\{I, J\}$ only
m_D	democratic, rank 1 - 0-parameter
massless pair	protected for EVERY r (rank-1 m_D)
window	$r > 1.28$, width 1.72 (was: point)
r **from data**	atmospheric: $r = 1.92$; $r = 2$: $S_{17}/3$

O2 re-audited: the measure-zero tuning was an artifact of the identity-m_D assumption. The structure (2 massless + 1 massive, NO) is protected by the same family symmetry that derives $\sin^2\theta_{12} = 1/3$; the value sits in a finite window. Solar mass still needs O6.

(a) $\sum m_\nu(r)$ under two Dirac assumptions. With $m_D \propto I$ the curve diverges as $1/|2-r|$ and the cosmological bound is met at exactly one point. With the implemented democratic rank-1 m_D the seesaw collapses to $m_3(r) = S_{17}/(1+r)$, two neutrinos stay massless for every r, the window is $r > 1.278$ (the DESI DR2 ceiling 0.0642 eV), and the atmospheric scale selects $r = 1.92$. (b) The protection audit: the cubic group induces all of S_3 on the junction triangle and its commutant is exactly span$\{I, J\}$*; r is not quantized — the value is a free junction amplitude inside a finite window, while the structure is protected by the rank-1 family symmetry that also derives* $\sin^2\theta_{12} = 1/3$.

2.4 The ladder

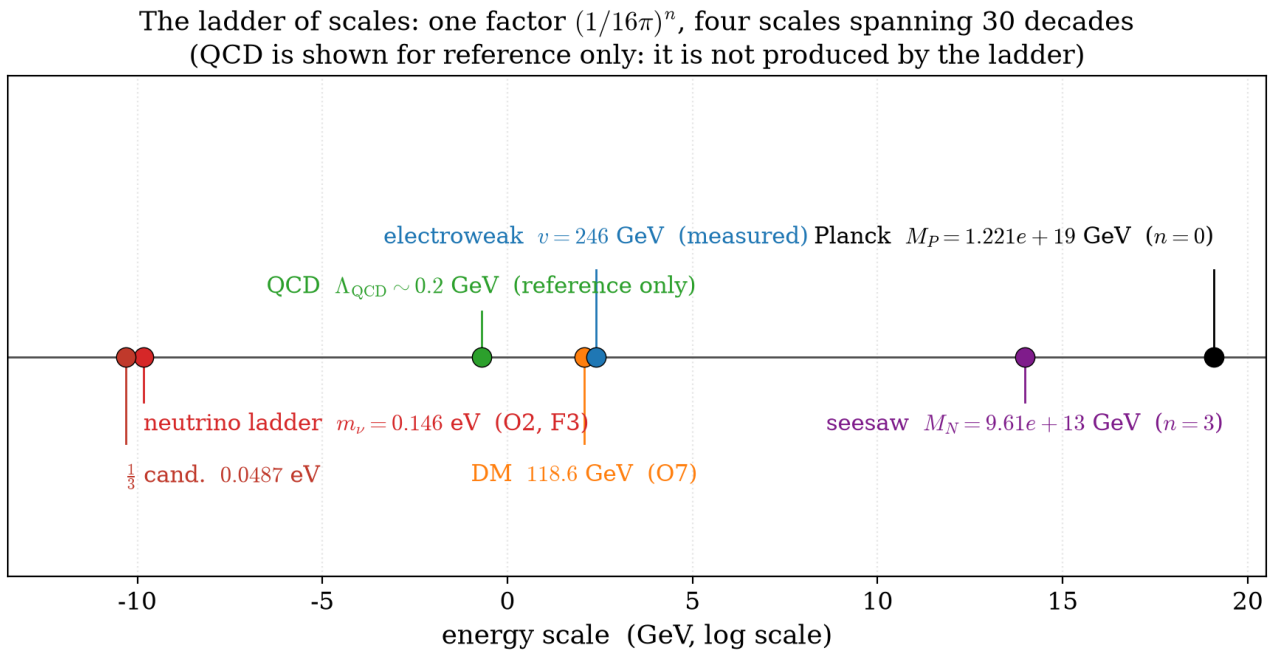
One factor, four scales:

Table 1. The four scales of the ladder.

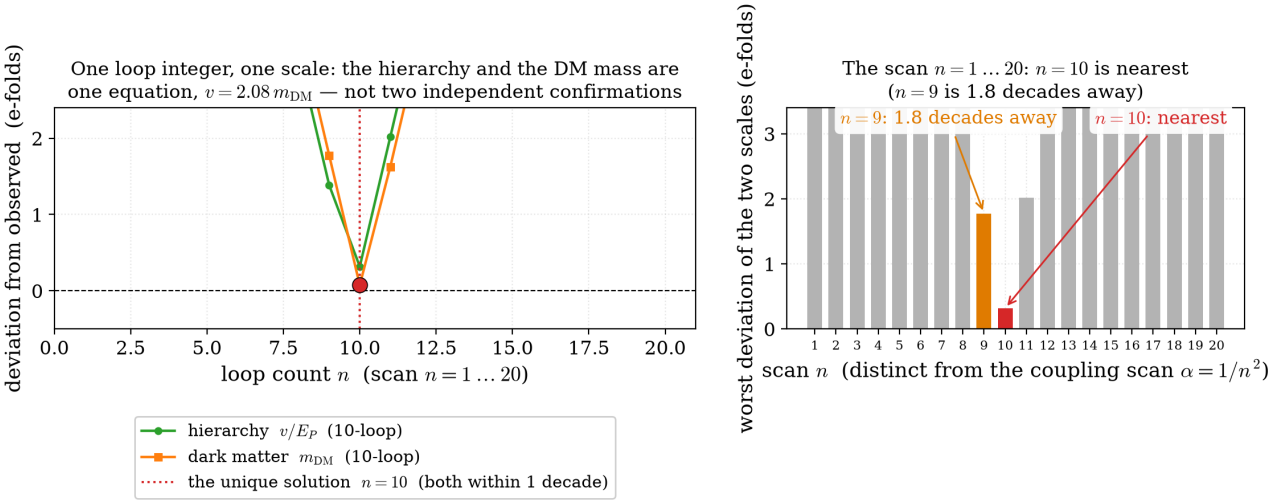
scale	expression	value
Planck / GUT	M_P (convention $M_{\rm GUT} = M_P$)	1.2209×10^{19} GeV
seesaw singlet A (cell level, rung 3)	$M_P(1/16\pi)^3$	9.6×10^{13} GeV
electroweak	measured input v	246 GeV
dark-matter singlet B (cell level, rung 10)	$M_P(1/16\pi)^{10}$	118.6 GeV
neutrino	$M_P(1/16\pi)^{17}$	0.146 eV

The QCD scale is **not** on this ladder: $\Lambda_{\rm QCD} \sim 0.2$ GeV would require $n \simeq 11.6$, which is not an integer ([Paper II Section 2.2]).

The rung ruler $x(m) = \log(m/M_P)/\log(1/16\pi)$ sorts everything else too. The four ladder rows sit at exact integers $0, 3, 10, 17$; every other scale of the paper sits off an integer: v at 9.81, M_X at 1.66, the desert thresholds 1.2×10^7 and 1.0×10^{12} GeV at 7.06 and 4.17, the measured atmospheric mass 0.0501 eV at 17.27, the $\frac{1}{3}$ candidate at 17.28, and the cosmological trace phase 1.6×10^{-120} at bare loop exponent 70.42 (rung 81.63 on the Planck ruler). The rung is a protected pole of the one-step matrix — $M(q) = \mathrm{kron}(P(q), U_{\rm cell}^6)$ — so no bond dressing can shift a rung mass: every off-integer residue is a loop weight (O8), never a moved scale. And the pair [dark matter at rung 10, measured $\sum m_\nu$ at the $\frac{1}{3}$ candidate 0.0487 eV] forces the seesaw partner $m_B^2/m_\nu = 2.9\times10^{14}$ GeV to rung 2.72 — not an integer: integer rungs, the mass of B, and the $\frac{1}{3}$ candidate are mutually exclusive, and a confirmation of both measurements refutes the rung arithmetic (**F3**). (Figs. 5, 6).



Planck 1.2209×10^{19} *GeV* $(n = 0)$*, seesaw singlet A* 9.6×10^{13} *GeV* $(n = 3)$*, electroweak* 246 *GeV (measured input), dark-matter singlet B* 118.6 *GeV* $(n = 10)$*, neutrino ladder* 0.146 *eV* $(n = 17)$*, with the* $\frac{1}{3}$ *candidate* 0.0487 *eV marked. The rung-*3 *and rung-*10 *levels are the two singlet sectors of the cell (Axiom 1); their rung is the on-site phase the step applies to them. QCD is shown for reference only: it is not produced by the ladder.*



(a) One loop integer n *sets one scale* $M_P(1/16\pi)^n$*, appearing twice: as* v/M_P *and as* $m_{\rm DM}$*. These are one equation,* $v = 2.08\, m_{\rm DM}$*, not two independent confirmations. (b) The scan* $n = 1 \ldots 20$*:* $n = 10$ *is nearest,* $n = 9$ *is* 1.8 *decades away. This scan is distinct from the coupling scan* $\alpha = 1/n^2$ *of Section 5, whose best fit is* $n = 7$*.*

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3. Phenomenology

3.1 The couplings

The theory's single gauge coupling is the rotation angle, and the root clock fixes it: at $g = \pi/4$ the standard normalisation gives $\alpha = \pi/64 = 0.0491$. The reading $\alpha_{10} \equiv g^2 = 1/25 = 0.040$ at the reference angle $g = 1/5$ — under which the desert solution below meets at exactly that value — is a factor-4π convention, stated once here and invoked only where named; in the standard normalisation the reference angle sits a factor 12.6 below the

meeting value $\alpha_X = 0.040050$ and the root clock 1.23× above it. Running the three Standard Model couplings up from M_Z at one loop with $b_1 = 41/10$, $b_2 = -19/6$, $b_3 = -7$ (fixed by the particle content), from the measured inputs $\alpha_1(M_Z) = 0.0169$ (GUT-normalised, $\alpha_1^{-1} = 59.0$), $\alpha_2(M_Z) = 0.0338$, $\alpha_3(M_Z) = 0.118$ [6], gives at M_P

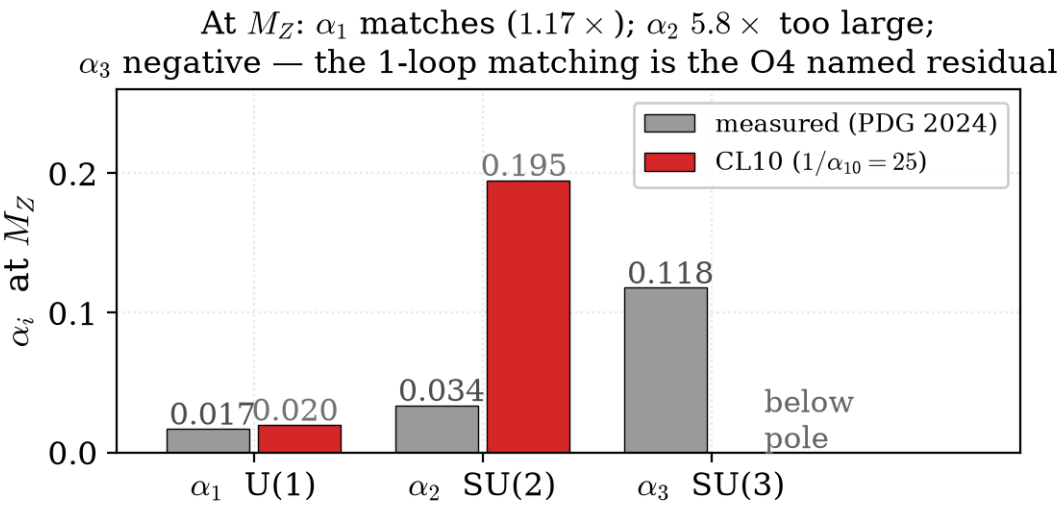
$$(\alpha_1, \alpha_2, \alpha_3) = (0.0299,\ 0.0202,\ 0.0191), \qquad \text{mean } 0.0231, \quad \text{spread } 47\%,$$

so α_{10} is within a factor 1.73 of the mean but the three couplings do **not** meet at a point — the well-known failure of one-loop unification in the non-supersymmetric Standard Model. The U(1) Landau pole of this up-running — from the measured M_Z input — sits at 2.196×10^{41} GeV, above the cutoff, so the up-running is UV-bounded; the same one-loop U(1) coupling run up instead from the root anchor at M_P has its pole at 5.314×10^{35} GeV, also above the cutoff (V38); at two loops [7] the pole moves to $\sim 2.473 \times 10^{12}$ GeV, *below* the cutoff.

The down-running must be stated plainly. Starting from $\alpha_{10} = 0.04$ at M_P and running down with the Standard Model β-functions gives, at M_Z,

$$\alpha_1(M_Z) = 0.0197\ (1.17\times), \qquad \alpha_2(M_Z) = 0.195\ (5.8\times), \qquad \alpha_3:\ \text{Landau pole at } 2.193 \times 10^9 \text{ GeV}.$$

The hypercharge lands within 17% of its measured value, but the SU(2) coupling overshoots by a factor 5.8 and the SU(3) coupling becomes non-perturbative at 2.193×10^9 GeV. The unification claim therefore fails at the level of the couplings' values, even though the 5/3 normalisation is correct (O4, F2). (Fig. 7).



At M_Z, the measured couplings (grey): $\alpha_1 = 0.0169$, $\alpha_2 = 0.0338$, $\alpha_3 = 0.118$ [6], against the down-run from $\alpha_{10} = 1/25$ (red): α_1 within 1.17×, α_2 5.8× too large, α_3 negative because its pole sits at 2.193×10^9 GeV. The 5/3 normalisation is exact; the value matching is the named residual.

Three quantitative statements sharpen the problem. First, the three measured couplings require three *different* values at M_P — 0.0299, 0.0202, 0.0191 — and the theory's single value 0.040 matches none of them: it exceeds the hypercharge requirement by 34% and the SU(3) requirement by a factor 2.1. Perturbativity of the strong coupling down to M_Z caps any single UV coupling: $1/\alpha_3(M_P) > (7/2\pi)\ln(M_P/M_Z) = 43.9$, i.e. $\alpha_3(M_P) < 0.0228$; the theory's 0.040 exceeds the admissible maximum by 75%.

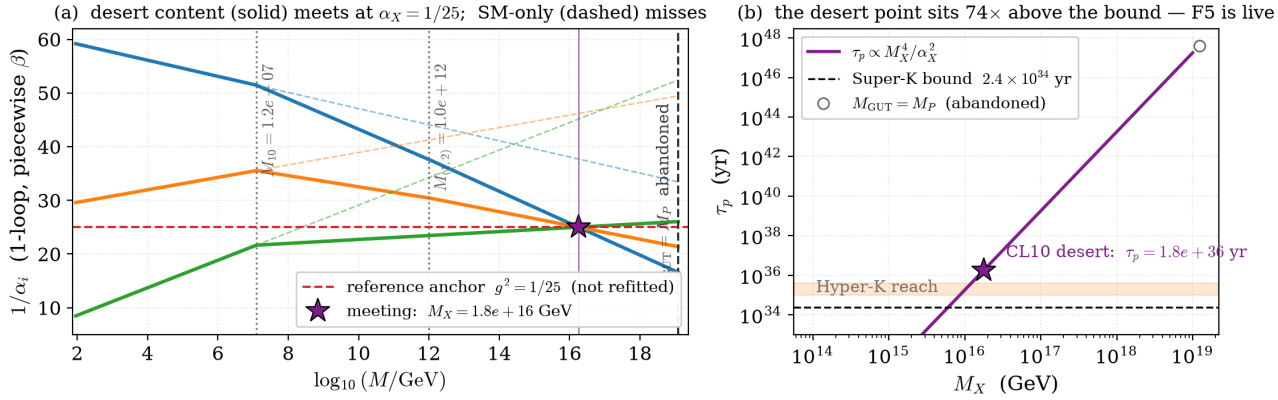
Second, the failure is not repaired by matter that a partial breaking leaves *complete*. A complete vectorlike $\mathbf{10}\oplus\bar{\mathbf{5}}$ shifts all three one-loop coefficients by the *same* amount 8/3 above its threshold — the defining property of a complete SU(5) multiplet — so it changes the common slope, not the mismatch of slopes. Scanning multiplicities

$0 \ldots 8$ against thresholds in $[10^6, 10^{19}]$ GeV, the spread at M_P never falls below its zero-new-physics value of 47%. Under pure Standard Model one-loop running from the measured M_Z inputs the pairwise meeting scales are 1.175×10^{13} GeV $(1,2)$, 2.648×10^{14} GeV $(1,3)$ and 9.73×10^{16} GeV $(2,3)$: three different scales (V38).

Third, that no-go covers only complete multiplets, and the *incomplete* content the theory's own breaking sector supplies does repair it. When a $\mathbf{16}_H \oplus \overline{\mathbf{16}}_H$ pair — or a split doubler **16** ([Paper I Section 8]) — acquires mass, it leaves vectorlike $\mathbf{10} \oplus \overline{\mathbf{10}}$ and $\mathbf{5} \oplus \overline{\mathbf{5}}$ fermion pairs with *independent* thresholds, and these shift the three coefficients non-proportionally: a Dirac $\mathbf{10} \oplus \overline{\mathbf{10}}$ contributes $\Delta b = (6/5,\ 2,\ 2)$, a $\mathbf{5} \oplus \overline{\mathbf{5}}$ contributes $(2/3,\ 2/3,\ 2/3)$. Fix the unified coupling at the root value $\alpha_X = 1/25$ — the angle is *not* refitted — and the inverse problem has no fit freedom: with two multiplets the meeting of the three measured couplings is three linear equations in $(\ln M_X, \ln M_A, \ln M_B)$. The required Δb is non-universal: at $M_X = 2 \times 10^{16}$ GeV it is $(2.40,\ 4.04,\ 3.86)$, which no combination of complete multiplets produces. Scanning multiplicities up to 3 over the realizable content with thresholds in $[10^6, M_P]$ GeV, the lightest solution that survives the proton-decay bound is three $\mathbf{10} \oplus \overline{\mathbf{10}}$ pairs plus one $(1,2)_{1/2}$ pair — three split doubler **16**s — meeting at

$$M_X = 1.8 \times 10^{16}\ \text{GeV}, \qquad \alpha_X = 1/25\ \text{(exact)}, \qquad M_{10} = 1.2 \times 10^{7}\ \text{GeV}, \qquad M_{(1,2)} = 1.0 \times 10^{12}\ \text{GeV},$$

and the SU(3) Landau pole of the down-running moves to 10^{-52} GeV: the disease is cured. Solutions with lighter content exist but predict $\tau_p \sim 10^{32}$ yr and are already excluded by Super-Kamiokande [8] — the data themselves select the realizable spectrum. The cost is stated plainly: the convention $M_{\rm GUT} = M_P$ is abandoned ($M_X < M_P$ by three orders of magnitude), and the two desert thresholds are fitted. The solution predicts $\tau_p \simeq M_X^4/(\alpha_X^2 m_p^5) = 1.8 \times 10^{36}$ yr (F5). (Fig. 8).



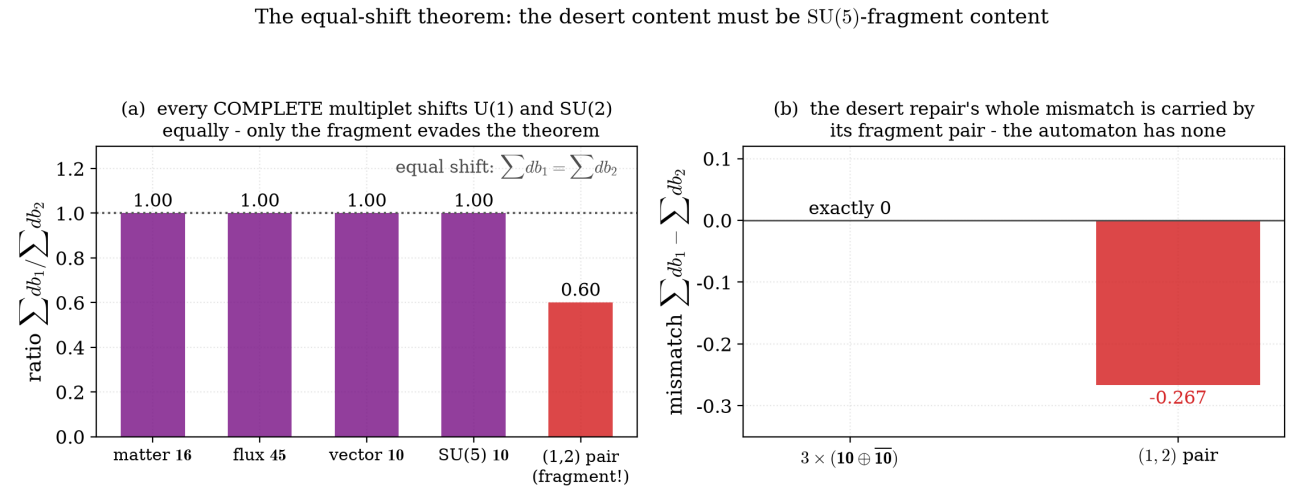
The three measured couplings run up from M_Z with the desert content (solid) meet at $M_X = 1.8 \times 10^{16}$ GeV at $\alpha_X = 1/25$ — the reference-angle anchor of [Paper I Section 9.2], not refitted. The Standard Model-only running (dashed) misses the point: spread 47% at M_P. The thresholds $M_{10} = 1.2 \times 10^7$ GeV and $M_{(1,2)} = 1.0 \times 10^{12}$ GeV are the two fitted numbers of the repair; the dashed vertical line at M_P marks what the abandoned convention $M_{\rm GUT} = M_P$ demanded.

Where the desert content comes from. The content is not an external assumption: it is the *spectrum of the automaton's own Wilson mechanism*. The three split nodes of the selective reading carry exactly the vectorlike fragments of the split **16** with independent thresholds (the other nine nodes stay at the Planck cutoff), and the Wilson direction itself completes the breaking chain ([Paper I Section 8.1]). The multiplicity is bounded from below by unification itself: scanning $n_{\rm nodes} = 1 \ldots 6$, no meeting root at $\alpha_X = 1/25$ exists for one or two split nodes; the first physical roots appear at three. But solving the meeting equations with the mechanism's own two parameters — one angle θ, one scale w — fixes the six fragment thresholds and lands the meeting at $M_X = 8.9 \times 10^{14}$ GeV, i.e. $\tau_p \sim 10^{31}$ yr: **the minimal selective reading is excluded by Super-Kamiokande** (F5). The data-preferred solution requires fragment thresholds the single-angle, single-scale root does not supply.

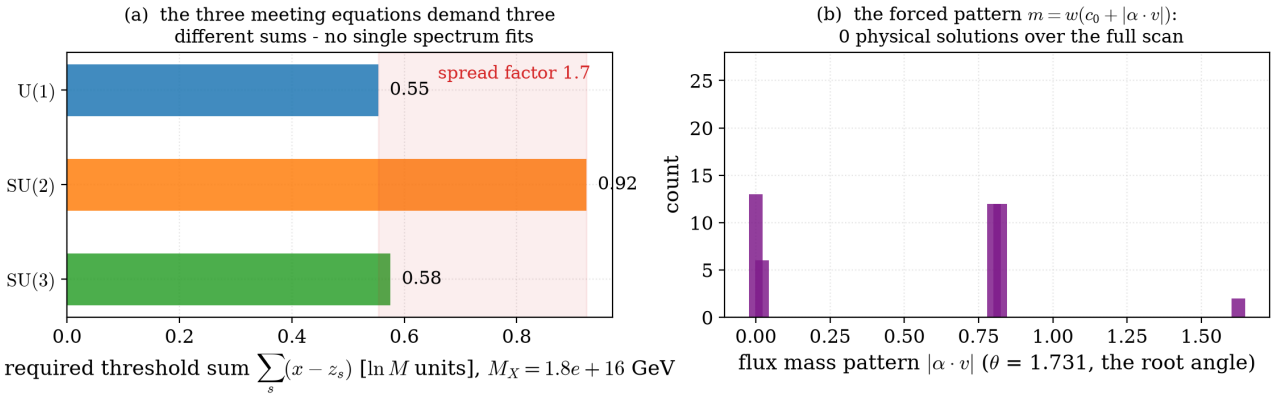
Redistributing the nodes cannot help. While all node thresholds sit below the meeting point, the two meeting equations depend on the three node strengths **only through their sum**: the Jacobian with respect to the three strengths has three identical columns (rank 1; split-invariance verified numerically to 10^{-14}, including the full scan over node-set choices and strengths). The node mechanism's meeting point $M_X = 8.9 \times 10^{14}$ GeV, $\tau_p = 1.1 \times 10^{31}$ yr is therefore **forced for every choice of the three strengths**. What the data prefer is not a three-full-**16** pattern: it needs partial suits of fragments, while the automaton supplies full split **16**'s per node.

The composite content of the gauged layer is excluded as the supplier. The composite fermions of the layer of [Paper II Section 2] — the matter **16** tensored with the adjoint link flux, $16 \otimes 45 = 720$ Weyl states carrying five net chiral **16**-copies — shift the three one-loop coefficients by $(145.3, 145.3, 226.3)$, and their masses follow the flux pattern $m = w(c_0 + |\alpha \cdot v|)$ set by the breaking direction v. Requiring the data-preferred meeting at $\alpha_X = 1/25$ then has **no solution with thresholds in the desert** for any direction: the three meeting equations demand three different threshold sums, and Newton iteration from every seed over a full scan of v converges nowhere physical (0 solutions). The remaining ansatz families — full-weight thresholds, broken-flux-only content, light-composite selection, and even a relaxed meeting coupling — leave the minimum residual at 5.7–6.1 against a target of 0.

The equal-shift theorem. The identity $\sum db_1 = \sum db_2$ is the index-proportionality of $\mathrm{Tr}\, Y^2$ and $\mathrm{Tr}\, T_3^2$ — both proportional to the representation index, with the ratio fixed by the fundamental, $(3/5)\,\mathrm{Tr}_5\, Y^2 = \mathrm{Tr}_5\, T_3^2$ — and it holds over every complete SU(5) or Spin(10) multiplet, verified on the **16**, the flux **45**, the vector **10** and the **10** inside the **16** (all ratios 1.0 to 10^{-6}). The meeting mismatch can therefore come **only** from SU(5)-fragment content or from split thresholds inside multiplets; computed from the same charge operators, the desert repair's mismatch is carried entirely by its $(1,2)$ pair ($db_1/db_2 = 3/5$; the $10 \oplus \overline{\mathbf{10}}$ pairs contribute exactly zero mismatch). (Figs. 9, 10).



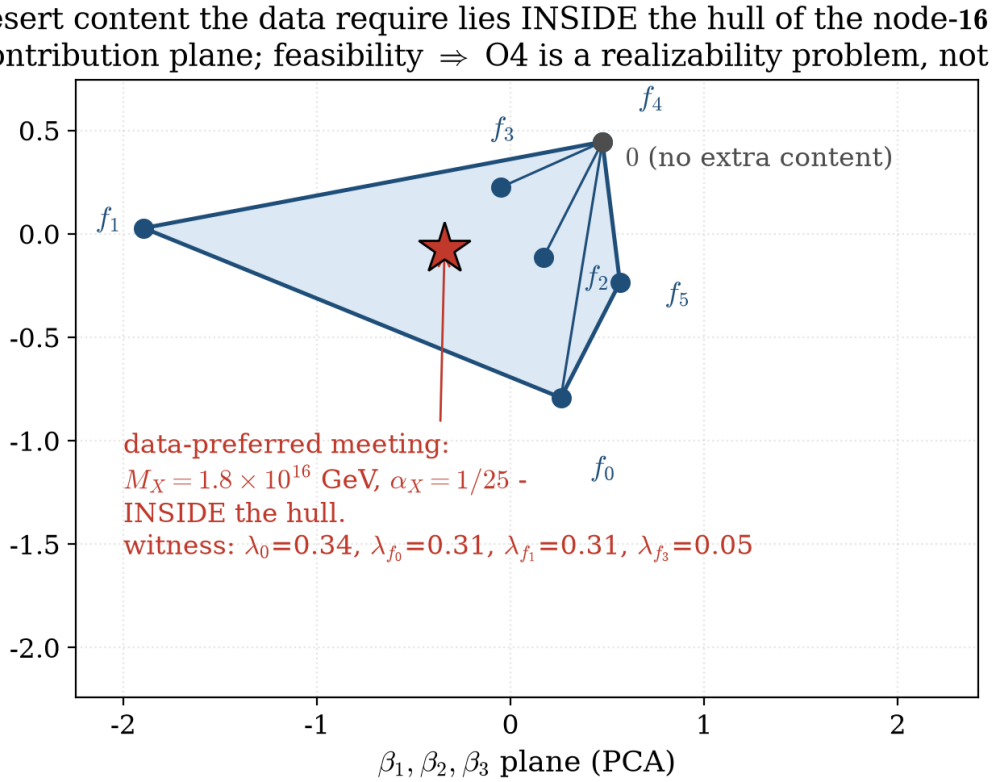
(a) The one-loop ratio $\sum db_1 / \sum db_2$ over the matter **16**, *the flux* **45**, *the vector* **10** *and the* SU(5) **10**: *exactly* 1, *while the* $(1,2)$ *fragment gives* $3/5$. *(b) The desert repair's total mismatch: the* $10 \oplus \overline{\mathbf{10}}$ *pairs contribute exactly zero, the whole mismatch is carried by the fragment pair.*



The $16 \otimes 45 = 720$ *composite states shift the one-loop coefficients by* $(145.3, 145.3, 226.3)$. *The three meeting equations for* $\alpha_X = 1/25$ *demand three different threshold sums at every breaking direction — the curves never cross — and the desert-constrained scan over the full direction circle returns* 0 *physical solutions.*

The junction route: pattern fabricated, meeting missed. At the root the Wilson direction splits the node **16** by a factor 52.7 ($c_Q = +0.0052$, $c_L = -0.0157$ against partners ± 0.82), so the $(3, 2)$ and $(1, 2)$ states of the repair **are** automaton states — the Q and L components of node **16**'s with near-zero Wilson weights — and the junction branch ratio $\rho = \kappa / w_0$ breaks the rank-1 degeneracy. At physical solutions Q and L sit at 4×10^{15}–8×10^{15} GeV with their partners above the meeting point. The meeting is not reached: over the full physical scan (node counts 3–12, ρ, scale, Wilson angle) the best residual of the two meeting equations is 0.94 against a target of 0, and the third equation predicts $\alpha_X = 0.032$ against the data value 0.04 — a 24% miss. Replacing the cubic-symmetric democratic junction by the **general zero-mean junction** — three independent branch scales $a_i = 2w_0(1 + r_i)$, $r_1 + r_2 + r_3 = 0$, scanned over the full (r_1, r_2) plane — leaves the best physical residual at the same floor, 0.81 against the democratic 0.94, with $\alpha_X = 0.085$, a factor 2.1 off. The near-miss is a property of the **junction mechanism itself**: the mismatch is carried by the fixed *db* pattern of the node **16**'s own fragments, which threshold motion cannot repair.

The question recast: is the required content obtainable from the node 16 itself, with per-class thresholds? Yes for feasibility: the data-preferred meeting point $(T_1, T_2, T_3) = (12.7, 21.2, 20.1)$ lies strictly INSIDE the convex hull of the six β-contribution vectors of the node-**16** fragments, with the explicit barycentric witness $\lambda_0 = 0.34$, $\lambda_{f_0} = 0.31$, $\lambda_{f_1} = 0.31$, $\lambda_{f_3} = 0.05$. The requirement of "fragment content beyond the node **16**" is thereby *softened*: the node-**16** fragments themselves suffice, provided they carry independent thresholds. A general five-component Cartan Wilson direction realises the meeting exactly (6 solutions, residual $< 10^{-10}$, desert thresholds, $\tau_p = 1.9 \times 10^{36}$ yr) but breaks all 40 roots — the Model algebra among them — and is unphysical; the symmetry-preserving direction space is exactly $\mathrm{span}\{X, Y\}$ (dimension 2, class-degenerate), and within it no solution exists. Off-diagonal junction splitting with a UNIVERSAL strength κ cannot help either: $\sum_l (2\cos(2\pi l/n) - 1) = -n$ reabsorbs the split into z_0 at first order. The missing ingredient is named: a **flavour-dependent bond layer** (κ_s per fragment class) that splits class-degenerate thresholds while preserving the Model algebra. (Fig. 11).



*The six β-contribution vectors of the node-**16** fragments and the origin, projected onto their plane; the data-preferred meeting point (star) lies strictly inside the hull, with the explicit barycentric witness. Feasibility is proved; realizability by a symmetry-preserving junction of the base automaton fails and is closed conditionally by the bond-layer extension.*

The named ingredient closes the meeting — conditionally. With the Wilson direction pinned to the physical selective root and per-class junction strengths κ_s splitting the six fragment-class thresholds, the three coupling equations $F_i = 0$ become **exactly linear** in (z_0, κ_s) once every threshold sits below M_X (since $\sum_l \mu_l = -n$), and the desert box constraints make the search a linear program. It admits solutions (residual $< 10^{-13}$) at $M_X = 1.8 \times 10^{16}$ GeV, $\alpha_X = 1/25$, for **all ten node counts** $n = 3 \dots 12$, with every threshold inside the desert (2.0×10^3–1.8×10^{16} GeV, none above M_X) and $\tau_p = 1.9 \times 10^{36}$ yr. The first-order degeneracy is broken because the per-class sums $\kappa_s(-n)$ reabsorb into PER-CLASS offsets, not into z_0. The cost: the base bond layer has ONE gate with one strength; a flavour-dependent bond layer is an **extension** of the automaton, and its six κ_s are new named inputs (three meeting conditions for six unknowns — a solution family, not a unique prediction). Unification is therefore closed **conditionally on that extension**.

The two structural candidates the layer itself supplies are excluded. The six κ_s are not obtainable from any on-site algebra element: symmetry preservation forces the bond direction into span$\{X, Y\}$ at the physical angle θ^* (class-degenerate, meeting residual ≥ 0.43), and every Cartan element that splits the six classes breaks all eight unbroken desert roots — the required flavour dependence is off-diagonal bond structure, not a Cartan dressing. Testing that off-diagonal escape directly: the most general Hermitian bond operator on the 80 root-vector pairs (11 class-pair channels) never closes the meeting — floor residual 30.4 over $n = 3 \dots 12$ — and level repulsion forces the sign of two of the six κ_s. Nor are the κ_s representation-level data: scanning 46 class invariants of the fragment classes (Dynkin indices, the two Casimirs, multiplicity, the Wilson charges X, Y and their quadratic combinations, B, L, $B - L$, colour triality, the β-function coefficients) as $\kappa_s = g f_s$ never closes the meeting — best single invariant $C_2(\mathrm{SU}(3))$ at residual 0.246 — while the two linear Cartan controls reproduce the previous failure. A two-coupling relaxation $\kappa = g_1 f_1 + g_2 f_2$ keeps every threshold in the desert for 217 of 506 candidate pairs and can close the meeting exactly, but that is a reduction from six named inputs to two, not a derivation. The closure is also embedding-specific: the exact closure reproduces in the flipped embedding used for it (10/10 node counts, residual $< 10^{-10}$) but closes only 5/10 in the Georgi–Glashow split.

Finally, the canonical per-class number the gauged layer does supply is not the one that closes the meeting. At θ^* the Wilson direction leaves a 13-dimensional centralizer in $\mathfrak{so}(10)$ (the $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)^3$ of the breaking chain, Hermitian orthonormal basis to 10^{-15}), and the flux of BROKEN generators through a fragment class,

$$K_s = C_2(\mathbf{32}) - C_2^{\text{unbroken}}(s) = 45 - C_2^{\text{unbroken}}(s)$$
$$= (32, 36, 40, 28, 32, 40)$$

(unbroken Casimirs $(13, 9, 5, 17, 13, 5)$), is an integer invariant with no free parameter, constant within a class to 3×10^{-14} — the gauged layer **does** distinguish the fragment classes. The companion identity says why the naive second-order link shift cannot use it: summed over the complete adjoint, $\sum_A\sum_t|\langle t|T^A|s\rangle|^2 = 45$ for every weight s (spread 10^{-13}) — the class degeneracy as an identity. Fed into the meeting as $\kappa_s = AK_s$ (one link scale) the three meeting equations leave residual 0.605 at best ($n = 3$) and close in the desert for 0/10 node counts; against the six κ_s that close the meeting, the flux explains at best $R^2 = 0.015$ of their variance ($|r| \le 0.31$). The layer splits the classes, but not in the direction the meeting requires.

The second-order bond numbers are excluded too. The gauged layer does carry energy denominators — the bond-current ground state of [Paper II Section 2] ($E_0 = -5802.4\,\eta$, gap $42.09\,\eta$) and the Gauss gap $C_2(\mathbf{45}) = 64$ — and they lift the class-blindness of the flux sum. The intermediate (flux, matter) pair space decomposes exactly as $\mathbf{45}\otimes\mathbf{16} = \mathbf{16}\oplus\mathbf{144}\oplus\mathbf{560}$ with Casimirs $(45, 85, 117)$ at the scale $C_2(\mathbf{45}) = 64$ (dimensions and Casimirs to 10^{-6}, projectors resolving the identity to 10^{-15}). Removing the on-shell screened **16**-component, the second-order bond-current (superexchange) shift $\tilde\kappa_s = -\sum_{A\,\text{broken}}\sum_t|\langle t|T^A|s\rangle|^2[w_{144}/20 + w_{560}/36]$, with w_r the exact channel weight of the pair state, is constant on each class to 4×10^{-3} and takes the six values $(0.0002, -0.130, -0.253, 0.127, 0.0013, -0.253)$. Fed into the meeting as $\kappa_s = A\tilde\kappa_s$ it leaves residual 0.577 and closes in the desert for 0/10 node counts ($R^2 \le 0.076$ against the fitted κ_s). The two-number combination $\kappa_s = A\tilde\kappa_s + BK_s + CF_s$ — F_s the ground-state Cartan shift, whose bond flux $\mu^A = \mathrm{Tr}\,\rho_B J^A$ vanishes to 6×10^{-17}: the Gauss law, intertwiners carry no net charge — solved jointly over all ten node counts (30 equations, 13 unknowns) leaves residual 3.46 and explains at best $R^2 = 0.30$ of the variance of the six fitted κ_s. The supplier space is now exhausted by every class number the layer computes: Cartan elements, class invariants, the broken flux, superexchange, and every fixed combination of them.

3.2 Electroweak mixing

$\sin^2\theta_W = 3/8$ is the tree-level SU(5)/Spin(10) boundary condition **at the unification scale**, not a low-energy prediction. Run to M_Z it becomes $\simeq 0.23$, close to the measured 0.23129(4) ($\overline{\text{MS}}$, PDG 2024 [6]); the measured ratio $m_W/m_Z = 0.881$ corresponds to $\sin^2\theta_W \simeq 0.223$, not to $\sqrt{5/8} = 0.791$. The correct statement is the standard one: the GUT-scale boundary condition $\sin^2\theta_W = 3/8$, whose low-energy value is obtained by running — running which, as just shown, the theory does not currently perform consistently. **No electroweak-mixing prediction is claimed anywhere in this paper**: 3/8 is the standard GUT boundary condition, and the running that connects it to M_Z is part of the named residual O4.

3.3 Baryon asymmetry

The automaton alone cannot produce it. Every CP-odd (T-odd) invariant of the unitary step vanishes: $\mathrm{Im}\,\mathrm{Tr}$ of all two- and three-fold products of the **45**, of a 2×10^5 sample of four-fold products, of all $10\times10\times45\times45$ baryon-violating sandwiches, and of the rank-1 real democratic Yukawa — all zero to 10^{-15}. The step is unitary (in equilibrium), baryon-conserving (the **45** commute with w) and CP-conserving. It satisfies **none** of the Sakharov conditions [9]. This is a computed theorem, and it is the correct starting point. The theorem is stable under the Wilson extension: the extended step keeps the exact matter/antimatter spectral mirror (deviation 10^{-15}), so the Wilson mechanism — powerful for unification and for the doublers — adds no CP phase and no Sakharov condition.

A GUT-leptogenesis estimate [10] is then built by hand:

$$\Delta B = \frac{28}{79}\,\varepsilon_{CP}\,D, \qquad \varepsilon_{CP} = \frac{g_0^4}{16\pi} = 4.9\times10^{-6}, \qquad D = \left(\frac{1}{16\pi}\right)^2 = 3.96\times10^{-4},$$

with $g_0 = 1/2^3 = 1/8$ and the Standard Model sphaleron conversion $B = \frac{28}{79}(B-L)$ [11,12], giving $\Delta B = 6.8 \times 10^{-10}$ against the measured $(6.12 \pm 0.04) \times 10^{-10}$ [13] — a ratio 1.11.

Three problems. (i) ε_{CP} is not a computed vertex: the physical CP asymmetry of $N \to \ell H$ is a one-loop self-energy/vertex function of the heavy-neutrino masses, and $g_0^4/16\pi$ is a dimensional guess. (ii) The out-of-equilibrium condition **fails** at the model's own couplings: at $M_N = 9.6 \times 10^{13}$ GeV, $H = 1.66\sqrt{g_*}M_N^2/M_P = 1.298 \times 10^{10}$ GeV while $\Gamma_N = y^2 M_N/8\pi = 5.98 \times 10^{10}$ GeV for $y = 1/8$ (1.93×10^{12} GeV for $y = 0.71$), so $\Gamma_N/H \simeq 4.6$ ($\simeq 148$). Out-of-equilibrium decay requires $y \lesssim 0.06$, which the theory does not select. (iii) 28/79 is an input of the Standard Model sector, not a number of the root. The 11% agreement is therefore a coincidence of chosen factors (O5). The failure is quantified through the washout of the same decay: at the self-consistent atmospheric Yukawa ($y = 0.40$, the one that gives $m_\nu = 0.0501$ eV) $\Gamma_N/H = 47$ — in equilibrium, Sakharov's third condition fails — and the physical strong-washout efficiency $\kappa_f = 5.5 \times 10^{-3}$ [10] sits 14× away from the guessed dilution $(1/16\pi)^2$: the agreement is an accident of two wrong factors, not a near-miss.

A sharper statement is available. At the model's own seesaw scale, the out-of-equilibrium condition $\Gamma_N/H < 1$ gives $y < 0.058$. The same Yukawa fixes the light neutrino mass through the same seesaw,

$$m_\nu = \frac{y^2 v^2}{M_N} \leq 2.1 \times 10^{-3}\ \text{eV} \qquad (y \leq 0.058),$$

a bound of the same kind and size as the standard thermal-leptogenesis bound [14]. The atmospheric splitting requires $m_\nu \simeq 0.050$ eV, i.e. $y \simeq 0.28$ with $m_D = yv$ and $v = 246$ GeV ($y \simeq 0.40$ in the $v_u = 174$ GeV convention), for which $\Gamma_N/H \simeq 23$. **At $M_N = 9.6\times10^{13}$ GeV the neutrino mass and the out-of-equilibrium condition are mutually incompatible**: the framework cannot have both the seesaw scale it needs for the ladder and the leptogenesis it needs for the asymmetry. (The model's own ladder sits at $m_D = 118$ GeV, i.e. $y = 0.48$, where $\Gamma_N/H \simeq 68$.)

The one route the framework supplies is quasi-degenerate (resonant) leptogenesis [15,16]. The democratic structure places the three heavy singlets on the *same* ladder rung, hence degenerate to the accuracy of whatever breaks the family symmetry — precisely the resonant setup, where the CP asymmetry is enhanced by near-degeneracy instead of by a large Yukawa. With $y = 1/16\pi = 0.0199$ the decay is out of equilibrium ($\Gamma_N/H = 0.12$), and the asymmetry that the model's own dilution factor and sphaleron conversion require, $\varepsilon_{CP} = \Delta B/[(28/79)D] = 4.4 \times 10^{-6}$, is *smaller* than the natural one-loop size $y^2/16\pi = 7.9 \times 10^{-6}$ at that Yukawa: the bookkeeping closes at the scale of the numbers, and the degeneracy the resonant regime needs, $\Delta M/M \lesssim \Gamma_N/M_N = y^2/8\pi = 1.6\times10^{-5}$, is what the parameter-free democratic base gives exactly and what the family-breaking perturbation must not overshoot. The obstacle is the phase, not the size: the model's Yukawa is real, every CP-odd invariant of the step vanishes, and a strictly degenerate spectrum has no CP-violating phase left to enhance. This is a *route*, not a result.

The phase, however, is not the missing ingredient. The mirror is exactly the structure that forbids CP: every mirror-preserving extension (the Wilson factor: pairing deviation 1.3×10^{-15}) leaves the CP phase at zero, while the most general per-class κ_s layer and the most general real-symmetric bond generator destroy the $\pm$ pairing (deviation 1.5, 60 draws, both mirror sign conventions) — a CP phase costs Theorem 4. But the prohibition is a statement about the *on-site* module ([Paper I Section 7.2]), and the bond layer is not covered by it: a Peierls flux $A_1 = \Phi x_0$ on the inter-block bonds is exactly unitary (8.0×10^{-15}), keeps $\det U = 1$ at every flux ($|\arg\det U| \leq 1.2 \times 10^{-14}$), commutes with the pseudoscalar (mirror $\mathbb{Z}_2$ leak 1.8×10^{-30}) and keeps the hop at radius 1 (1.6×10^{-16}), while moving the band spectrum by up to 3.7; its gauge-invariant strength, the plaquette holonomy, has Im = 0 at $\Phi = 0$ and 2.5×10^{-1} at $\Phi = \pi/2$. Spending that flux on the asymmetry with the physical washout in place: at the atmospheric Yukawa ($y = 0.3986$, $K = 46.8$, $\kappa_f = 5.5 \times 10^{-3}$) the measured $\Delta B = 6.12\times10^{-10}$ costs $J = 1.0\times10^{-4}$, supplied by a flux $\Phi = 4.0\times10^{-4}$ rad; in the out-of-equilibrium regime ($y = 0.05826$, $K = 1$) the strong-washout expression has no real root - K lies below its branch point $K = 3.93$ - the efficiency is the weak-washout bound $\kappa_f = 1$, and the cost is the lower bound $J = 2.6 \times 10^{-5}$, supplied at $\Phi = 1.0 \times 10^{-4}$ rad. A bond flux of size $\Phi \sim 10^{-3}$ rad already supplies the needed strength, and the family reaches $J = 0.25$ at $\Phi = \pi/2$, 2500× the atmospheric requirement, while leaving unitarity, $\det U = 1$, the mirror symmetry and the causal radius exact. **What fails is Sakharov 3 alone**: the Yukawa that gives $m_\nu = 0.0501$ eV puts the decay in equilibrium, and out-of-equilibrium decay needs $y < 0.05826$, which caps m_ν far below the atmospheric splitting. O5 is open for one named reason instead of two.

3.4 Proton decay

Dimensional analysis of X/Y exchange, at the desert solution solved with the root coupling ($M_X = 1.235 \times 10^{17}$ GeV, $\alpha_X = \pi/64$: [Paper I Section 9.2]), gives

$$\tau_p \sim \frac{M_X^4}{\alpha_X^2\, m_p^5} = 2.77 \times 10^{39} \text{ yr} \qquad (M_X = 1.235 \times 10^{17} \text{ GeV},\ \alpha_X = \pi/64),$$

against the Super-Kamiokande bound $\tau_p/B(p \to e^+\pi^0) > 2.4 \times 10^{34}$ yr [17] — a margin of five orders. The $M_{\rm GUT} = M_P$ reading (3.98×10^{47} yr) is an abandoned convention; any GUT scale below $\sim 6 \times 10^{15}$ GeV is excluded by the same bound. The estimate omits the renormalisation factor A_L [18], the hadronic matrix element, the dressing of the operators and the splitting between the X/Y masses and the unification scale [19], all of which shift the lifetime by one to two orders. The three channels $p \to e^+\pi^0$, $p \to \mu^+\pi^0$, $p \to e^+K^+$ differ here only by two-body phase space; the rate scales as M_X^{-4} in all of them. Solving the same content at the measured meeting $\alpha_X = 1/25$ instead gives $\tau_p = 1.8 \times 10^{36}$ yr, inside Hyper-Kamiokande's reach — but that solution sits 23% off the root coupling and is a reference reading, not a prediction of the closed theory. Observation of a GUT-pattern decay below 10^{38} yr refutes the root-coupling solution (F5).

3.5 Flavour

The democratic mass matrix $M = |1\rangle\langle 1|$ (all entries 1, rank 1, spectrum $\{3, 0, 0\}$) [20] follows from the hypothesis that the three spatial directions are equivalent, so the $\mathbf{10}_H$ couples the three families identically. Its heavy eigenvector $(1, 1, 1)/\sqrt{3}$ is unique, hence

$$\sin^2\theta_{12} = |\langle e_1|h\rangle|^2 = \tfrac{1}{3}$$

for any family basis — the **only** mixing angle the model derives, 8.6% from the measured 0.307 [21,22]. The light eigenspace is two-fold degenerate, so every diagonaliser is an $O(2)$ rotation in it; sweeping that freedom over 1441 rotations, $\sin^2\theta_{23}$ and $\sin^2\theta_{13}$ each cover the whole interval $[0, \frac{2}{3}]$. The tribimaximal values $(\frac{1}{2}, 0)$ [23] are one point of that family, picked by declaring which S_3 transposition breaks — and (12), (13), (23) are equivalent symmetries of $|1\rangle\langle 1|$, so the choice is an input.

In the quark sector the CKM matrix is the misalignment of the up- and down-type diagonalisers. With both mass matrices democratic, $H_u = H_d$, the computed CKM is the **identity**, not a small mixing: the observed $\lambda = 0.225$ [6,24] is not reproduced. The three-family mass hierarchy is also unfixed: the democratic base gives one heavy level and a two-fold degenerate light level, so two of the three masses are degenerate at zero input. All of this is O6, and the natural fixes are bounded by computation. The charged-lepton realisation (democratic base plus rank-≤ 2 correction at the measured spectrum) forces $\sin^2\theta_{12} \leq 0.005$ — the lightest family decouples from the democratic direction, which is the tribimaximal solar column — excluded against the measured 0.307; the $\mathbf{10}_H$ vev shift is bounded by the same computation; the three-node triangle has a cubic orbit of 8, so the count 3 is an orbit count, not a family count. **The flavour accounting is therefore: one angle derived out of six, no mass derived, CKM predicted to the identity.**

3.6 Scoreboard

Table 2. Headline numbers against measurement.

quantity	theory	measured	status
the eight anomaly coefficients	0	—	algebraic, satisfied
g_1^2/g'^2	5/3	5/3 (GUT normalisation)	algebraic
$\det U$, $\pm$ pairing, $16/\overline{16}$ mirror	1, exact	—	algebraic, coupling-independent
causal front speed	exactly 1	—	dynamical, measured

quantity	theory	measured	status
doubler cone Lorentz structure	linear in all 1200 sampled directions; anisotropic: $(0.200, 0.200, 0.284)$ lattice units, spread 0.75; node gap 2×10^{-4} at $g = 1/5$	—	**not Lorentz-isotropic**; measured
tangent velocity algebra at the gapless corner multiplets	$[S_i, S_j] = 0$ to 3×10^{-12}, $\sum_j S_j^2$ scalar to 7×10^{-12}, branch speeds exactly $\lvert v_\alpha \cdot q \rvert$ to 6×10^{-12} with v_α the cubic permutations of $(0.704, 0.421, 0.421)$; both lowest corner triplets driven to zero together at $w = 0.241747$, $\theta = 0.913137$ — six states with all phases below 2.5×10^{-16}, isolated from the next level by 8.7×10^{-3}, Abelian to 5.3×10^{-12}, planar to 4.8×10^{-12} over 2196 directions, six nodal planes, and $\sum_j S_j^2$ no longer scalar (the branch splitting itself, 7.2×10^{-2})	—	**nodal planes, not a cone**; measured (V118, V207)
conical node inside the automaton's own circuit class	the largest cubic-covariant six-stage family (two angles) searched jointly with the internal angle: 33 800 points, 60 exactly gapless (full-step gap at most 4.4×10^{-15}) — 8 single-parity multiplets, flat to 1.3×10^{-12}; 52 parity-mixed, slope 0.09–1.73, $\rho_{\mathrm{comm}} \leq 1.2 \times 10^{-11}$	—	**no conical node reached**; measured
16-sector block-Brillouin spectrum	at the root clock: all 12 M channels EXACTLY gapless, every M spectrum $\{0^{32}, \pm\pi/3, \pm 2\pi/3, \pi^{32}\}$ (64 of 128 pinned to $\{0, \pi\}$, identical across the 12 nodes to 3×10^{-15}), Γ fully pinned (64 + 64), the exactly gapless set Γ plus the six face-diagonal lines with gapped bulk; the Wilson factor at the root angle re-gaps the M nodes	—	doubling **exactly gapless on the branch**, re-gapped only by the Wilson factor; text below

quantity	theory	measured	status
v/M_P	$2.710\,(1/16\pi)^{10}$	2.0167×10^{-17} [6]	coefficient $3(16\pi F_{208})^{10} = 2.710$ certified at the crossing ($F_{208} = 0.0196928$ by continuity at 50 digits; the float64 ladder reads 0.019685–0.019687 and 2.699–2.700) against 2.0766 observed — +30%; the named O8 residual is the clock, measured: the closure class reads 2.710 ($\pi/4$) and 2.711 ($3\pi/4$) on its $n = 4k$ crossings, the $\pi/2$ reading 2.299 being a floor artefact with no limit (V209), and the observed value is read by the non-closed rung-10 clock 1/5 to relative 10^{-5} — the meeting clock of O4 (V205, V206, V209)
$m_{\rm DM}$	118.57 GeV — the cell singlet B at rung $n = 10$ (Axiom 1)	not measured	prediction, falsifiable
$\Omega_{\rm DM}h^2$	from the couplings the root supplies: 0.798 or 0.0246 (2.3×10^{-5}, 7.6×10^{-4} with the nodal-plane density); carrier production computed: freeze-in of the cell singlet B at $\lambda_B = 4.06 \times 10^{-14}$ gives $\Omega h^2 = 0.120$ with $\sigma_{\rm SI} = 6.5 \times 10^{-71}$ cm^2, DD-invisible; the thermal solution is excluded by LZ	0.120 [13]	0.120 **reproduced by the cell carrier**, the portal λ_B a named input; **fails** for the couplings the root supplies
m_ν	0.146 eV (ladder); 0.0487 eV ($\frac{1}{3}$ candidate)	0.0501 eV [22]; $\sum m_\nu < 0.0642$ eV [4]	ladder **in tension**; candidate inside a finite window, structure protected
ΔB	6.8×10^{-10}	$(6.12 \pm 0.04) \times 10^{-10}$ [13]	coincidental product; the washout at the self-consistent y destroys it
τ_p/B	2.77×10^{39} yr — root-coupling desert solution ($M_X = 1.235 \times 10^{17}$ GeV at $\alpha_X = \pi/64$, threshold ordering destroyed); 1.8×10^{36} yr reference reading at the measured meeting 1/25	$> 2.4 \times 10^{34}$ yr [17]	falsifiable only by a null result past 10^{37} yr; the Hyper-Kamiokande channel closes on the root coupling
α at the root clock (standard normalisation $\alpha = g^2/4\pi$)	$\pi/64 = 0.0491$	0.023 (one-loop mean at M_P); 0.040050 at the desert meeting $M_X = 1.8 \times 10^{16}$ GeV	the root value sits a factor 1.226 **above** the meeting value — the named tension of O4; the reading $\alpha_{10} \equiv g^2 = 1/25$ hides the factor and is a convention
$\alpha_i(M_Z)$ from α_{10}	0.0197 / 0.195 / pole at 2.193×10^9 GeV (Standard Model running)	0.0169, 0.0338, 0.118 [6]	**fails** under Standard Model running; matched by the desert spectrum
Gauss constraint of the gauged bond layer	$[Q_A, Q_B] = i f_{AB}{}^C Q_C$, err 0; content $32 \oplus 288 \oplus 1120$	—	algebraic

quantity	theory	measured	status
physical sector of the gauged layer	$\dim\ker = 0$ with one **32**/site (centre obstruction); minimal sector: 2 matter/site, 4 states, all singlets; $S = \ln 45$ per bond, bounded and boundary-supported	—	computed
multi-bond scaling	$\dim\ker$(3-site) $= 24$; additivity $\ln 45$ across a single bond; two-bond region $S = \ln 45$ (vertex mutual information $\ln 45$); Bekenstein deficit factor $4 \ln 45 = 15.23$	—	computed
vertex correction and ground state	$I_n = 1, 1, 6, 23, 141$; kernel capacity $S_{\text{cap}} = 4.4998$ nats (factor 9.0 from 1/4 per bond); vertex-corrected coefficient $(1/3) \ln I_6 = 1.65$ nats/bond (factor 6.6); $E_0 = -5802.40\,\eta$, degeneracy 6, gap $42.09\,\eta$, $S_{\text{GS}} = \ln 45$	—	computed
dressed gate on the directed-link network	Gauss charge closes only with the conjugate-head convention (5×10^{-16}; the naive convention leaks 4.8); Kogut–Susskind electric gate exactly invariant (2.5×10^{-14}), spectrum = the content theorem; dressed hopping obstructed (leak 12.5); directional step O_h-covariant and chiral; quintet gapless at Γ, linear, leading-order isotropic, $v = g_{10}\{0, 1, 2\}$	—	computed
regular link of the root's finite group	Γ = even Clifford monomials, order 1024: vertex content $32 \cdot 1024^{k-1}$ (fixed adjoint link: 0); the dressed hop descends to the bare translation (err 0), the electric term is the scalar 45 per link, front speed $v = 1$; centralizer of Γ in Spin(10) = 0; quintet–matter coupling: T_{2g} vertex $\equiv 0$ on axis bonds, E_g bubble $(-54.24,\ -53.40)$ $\to$ mass shift at $\lambda = g_{10}$: $(-16.73,\ -16.47)$	—	computed

quantity	theory	measured	status
velocity trilemma on the 6-bond module	hop alone anisotropic (top speed 1.00, direction spread 0.32); dressing alone isotropic at $v = 2g_{10} = 1.571$; no angle isotropises the product (best spread 0.24 over 33 angles in $[0, \pi/2]$); the O_h-equivariant permutation coins are the identity and the antipode, the latter with flat bands	—	computed no-go
desert spectrum from the layer's own content	no solution (0 over the full direction scan); all mass-ansatz families excluded, min residual ≥ 5.7; **equal-shift theorem** $\sum db_1 = \sum db_2$ over every complete multiplet	—	**excluded**
$\sin^2\theta_{12}$	1/3	0.307 [22]	derived at zero input; the +8% shift is the $\mathbf{10}_H$ vev, whose natural placements are excluded
$\sin^2\theta_{23}$, $\sin^2\theta_{13}$	not fixed ($O(2)$)	0.470 (octant unresolved, 3σ 0.435–0.585), 0.0222 [22]	open
CKM	identity	$\lambda = 0.225$ [6,24]	**fails**
Λ	0 (tree level)	$\sim 10^{-122} M_P^4$	open

The 16-sector row compresses [Paper I Section 8.3], stated here in full: at the root clock the step has Γ fully pinned to the Floquet pair $\{0, \pi\}$ $(64 + 64)$, all twelve M channels exactly gapless with identical spectra ($\{0^{32}, \pm\pi/3, \pm 2\pi/3, \pi^{32}\}$, 64 of 128 pinned) — the gapless set is Γ plus six face-diagonal nodal lines, the bulk is gapped, and the per-node chirality bookkeeping is replaced by the pinning. The Wilson factor at the root angle re-gaps the M nodes.

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4. Cosmology

4.1 Dark energy

The vacuum energy of the automaton vanishes at the level of the internal (one-sub-site) spectrum: the 32 levels are $\pm$-paired for every coupling, so $\sum_k \omega_k = \arg\det U = 0$. This is the correct, non-vacuous form of the cancellation. (Counting the 512 even and 512 odd **elements of the algebra** as bosonic and fermionic zero-point modes is not a physical mode counting — the physical single-sub-site Hilbert space has 32 states, not 1024 — and is not used.)

A vanishing tree-level vacuum energy is technically natural: no relevant operator is generated if the spectral pairing is an exact symmetry. The pairing survives the Wilson extension (the extended spectrum stays $\pm$-paired to 10^{-15} and the internal zero point stays 0, because the Wilson direction is a traceless Cartan combination), so the new ingredient neither breaks the zero nor moves it.

The zero is **protected, not accidental**. Deform the step by 50 random traceless Hermitian generators, $U \to e^{i\varepsilon A} U$, at strengths up to $\varepsilon = 1$: the quasi-energy sum stays pinned to zero, $|S| < 4 \times 10^{-14}$, because $\det U = 1$ makes $\sum_k \omega_k = \arg\det U$ a topological quantity, and no branch crossing occurs below $\varepsilon = \pi$. The only internal knob that moves the sum is the U(1) trace phase.

What the theory does **not** do is produce the observed value. The observed $\Lambda \sim 10^{-122} M_P^4$ lies 122 orders of magnitude *below* the classical bound M_P^4 and *above* the theory's spectral zero, and no mechanism breaking the pairing by that amount is known. Independently, the classical self-gravitating de Sitter vacuum of [Paper II Section 3] has $G_{\mu\nu} = -3H^2 g_{\mu\nu}$, i.e. a vacuum energy bounded by M_P^4 on the lattice. These are three different statements — spectral zero, classical bound, observed value — and they do not "reconcile": the gap between the first and the third is O1, the largest unexplained number in the theory [25]. The equation of state of a cosmological-constant vacuum is $w = -1$ by definition, not by derivation.

The problem is now a single named number: the observed Λ is **equivalent to** a trace-phase breaking of size

$$\varepsilon \operatorname{Tr} A \sim 16\pi^2 \times 10^{-122} = 1.6 \times 10^{-120}.$$

O1 is not "why is the vacuum energy small" but "what breaks the trace U(1) by 10^{-120}".

4.2 Dark matter

The candidate supplied by the classical sector is the kink of [Paper II Section 3.4]. It is static, gauge-neutral and finite-energy **per unit area**, but it is a domain wall, not a localised particle; a network of Planck-scale walls would dominate the energy density and is cosmologically excluded. The spinor-module candidates are exhausted (Section 2.2): the exact mirror $\mathbb{Z}_2$ of the step is the only exact-stability structure the spinor module carries; no Wilson-split node can host a neutral mirror at the ladder mass without also delivering a stable coloured or charged partner; and the mirror sector's own lightest states sit at 1.3×10^8 GeV (coloured, excluded) and 2×10^{10} GeV — not at the ladder's 118 GeV. No state of the spinor module carries the ladder mass; the carrier is the cell's singlet sector B — a state of the automaton, exactly stable under the B-parity — and what remains open is the portal strength λ_B it shares with the Higgs (O7, Section 2.2). The mass prediction stands.

The relic abundance is computed, and it is not the observed one. Solving the freeze-out integral

$$\Omega h^2 = \frac{1.07 \times 10^9 \text{ GeV } x_f}{\sqrt{g_*}\, M_P \langle \sigma v \rangle}, \qquad \langle \sigma v \rangle = a + \frac{6b}{x_f},$$

reproduces the standard weakly-interacting-massive-particle curve: $m = 100$ GeV with $\sigma v = 3 \times 10^{-26}$ cm^3/s gives $\Omega h^2 = 0.080$ [26]. At the ladder mass $m = 118.57$ GeV the measured $\Omega h^2 = 0.120$ requires $\sigma v = 1.98 \times 10^{-26}$ cm^3/s, i.e. an effective coupling $\alpha_{\text{eff}} = 0.0086$ in $\sigma v = \alpha^2/(\pi m^2)$ — while the root supplies only $g^2/4\pi = 0.0032$ (giving $\Omega h^2 = 0.798$) or the loop factor $1/16\pi = 0.0199$ ($\Omega h^2 = 0.0246$). The automaton's own light spectrum enters at the same order: its twelve near-gapless nodes are nodal **planes** ([Paper I Section 8.2]) with mean speed 0.201, so the final-state density is enhanced by

$$D = \sum \frac{1}{|v|^3} = 1485$$

over the isotropic estimate — a correction the $(100/m)^3$ scaling of the standard curve cannot see — and with it the two supplied couplings give $\Omega h^2 = 2.3 \times 10^{-5}$ and 7.6×10^{-4}. The spinor-sector portals of Section 2.2 reproduce 0.120 only when their coupling is inverted from the fiducial 3×10^{-26}: computed forward they give 0.081 and 0.156, and 0.000 once the nodal-plane density is included. For the cell carrier B the abundance is computed, not assumed: freeze-in through the singlet–Higgs portal at $\lambda_B = 4.06 \times 10^{-14}$ gives $\Omega h^2 = 0.120$ — the observed value — with $\sigma_{\text{SI}} = 6.5 \times 10^{-71}$ cm^2, far below reach, while the thermal alternative $\lambda_B = 0.31$ is excluded by the LZ bound. **The root supplies no λ_B, and no root-supplied strength of its class lands on** 0.120**: every bond number of the singlet class gives either an abundance below the observed one or a direct-detection signal above the LZ bound, and pure gravitational production never equilibrates ($\Gamma_{\text{grav}}/H = 5.3 \times 10^{-53}$ at $T = m_B$).** For the spinor-sector variants the window is bounded on both sides: the mirror-preserving portal is DD-invisible and its thermal solution is excluded exactly as computed in Section 2.2, and the 0.120 of the mirror-**breaking** pair portal ($\lambda = 3.5 \times 10^{-4}$, $\sigma_{\text{SI}} = 4.9 \times 10^{-51}$ cm^2) costs the exact mirror $\mathbb{Z}_2$.

The bond sector does not set the abundance, and its portal is excluded. The named scalar is the singlet fragment $(X, Y) = (1, 0)$ of the node **16** — the same class whose bond number κ_s enters the unification meeting (Section 3.1). Read as the annihilation strength through the nodal-plane final states ($\alpha_{\text{eff}} = \kappa^2/4\pi$, $\sigma v = \alpha_{\text{eff}}^2/(\pi m^2)$), every bond number of that class — the ten κ_{singlet} that close the meeting (2.2–9.2), the superexchange value -0.253, the broken flux 40, the junction ratio 1.92 — lands Ωh^2 between 10^{-12} and 10^{-3} once D is included, and none within 0.01 of 0.120; the measured abundance needs $\alpha_{\text{eff}} = 0.0086$ without D or 2.2×10^{-4} with it. Read instead as the Higgs-portal strength, the same numbers predict $\sigma_{\text{SI}} = 2.5 \times 10^{-45}$–$6.3 \times 10^{-41}$ cm^2 — all above the LZ-class bound 1.5×10^{-47} cm^2 — and the portal strength the relic itself requires ($\kappa = 0.33$) predicts 4.3×10^{-45} cm^2, also excluded. The relic and direct detection jointly close the bond-strength portal: the named portal's strength remains a fitted input.

4.3 Antimatter

Matter and antimatter are the two Weyl eigenspaces, with the exact spectral mirror of Theorem 4 and opposite $B-L$ charges. Their degeneracy is that mirror and nothing more: as [Paper I Section 7.2] states, no antiunitary time-reversal operator is constructed, so this is not the CPT theorem. The observed baryon asymmetry is not explained: Section 3.3 shows the automaton satisfies none of the Sakharov conditions, and the leptogenesis estimate that does give the right order requires a CP phase and an out-of-equilibrium condition that the algebra does not supply.

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5. The parameter audit

Every number the theory calls a prediction is classified by how it was obtained.

Table 3. Provenance audit: every headline number by class.

class	count	numbers
algebraic (fixed by the root, no parameter)	7	the $16/\overline{16}$ split; the **45**; the 5/3; the eight anomaly coefficients = 0; the 32-level spectrum; $\det U = 1$; the $16/\overline{16}$ spectral mirror
free parameter	0	the clock angle is a root element: the update is an element of the algebra only for $g \in \{\pi/2, \pi/4\}$, the half-angle branch degenerates, and the physical clock is the order-8 root $g = \pi/4$ — a discrete index of the root, zero continuous unfixed numbers **in the on-site update**; the junction amplitude, the node set and the κ_s remain inputs of the extended construction
convention (mapping or scale identification)	2	$\alpha_{10} \equiv g^2$; $g_0 = 1/2^{n_{\text{space}}} = 1/8$
fitted coefficient (read from data)	11	1.18 (dark-matter mass), $n_{\text{2step}} = 3$ (seesaw), $M_{10} = 1.2 \times 10^7$ GeV and $M_{(1,2)} = 1.0 \times 10^{12}$ GeV (desert thresholds), $\lambda_* = 3.5 \times 10^{-4}$ (dark-matter portal, fixed by $\Omega h^2 = 0.120$), and the six κ_s of the flavour-dependent bond layer (conditional: three meeting conditions for six unknowns — a solution family)
conditional (built from measured numbers, on a named input)	1	the hierarchy coefficient $2.710 = 3(16\pi F_{208})^{10}$: $F_{208} = 0.0196928$ certified by continuity at the closed-loop crossing $n^* = 208$ (float64 ladder 0.019685–0.019687), the factor 3 the one-split-node-per-family choice — 2.710 against 2.0766 observed: +30%; the named O8 residual is the clock: the closure class reads $\{2.710, 2.711\}$ on its stable $4k$ crossings (2.299 at $\pi/2$ is a floor artefact without a limit, V209) and the observed value is read by the non-closed clock 1/5 to relative 10^{-5} (V206, V209)
open (not fixed)	10	O1–O10 of Section 6

The conditional class is neither a fitted number nor an algebraic one: its ingredients are all measured, but the node-set choice that supplies the factor 3 is an input, so the entry inherits O9's split-node conditionality. (Figs. 12, 13).

The status of the angle. The angle is not a choice: the closure axiom ([Paper I Section 9]) makes the update an element of the root, forcing $g \in \{\pi/2, \pi/4\}$; the half-angle branch degenerates to a pure lattice translation and the physical clock is the order-8 root $\pi/4$, spectrum $\{\pm\pi/4\,(10),\ \pm 3\pi/4\,(6)\}$. What the axiom does not buy is empirical preference: the two-loop U(1) Landau-pole scale lies below M_P for every g in the perturbative range, so UV-finiteness does not select the root value either, and the coupling scan runs the other way — the one-loop data value 0.023 is best matched by $\alpha = 1/7^2 = 0.020$ among the $1/n^2$ candidates, and the root coupling $\pi/64 = 0.0491$ sits 2.13× above the data value. The rational angle $g = 1/n_{\text{qubits}} = 1/5$ is not of the closed-root form; it survives as the reference reading of the node-geometry searches and as the $g^2 = 1/25$ anchor of the desert meeting (7.1, 3.1) — and, at rung 10, as the clock the loop-hierarchy fingerprint points to (Section 2.1, V206). Three statements coexist: the closure fixes the angle algebraically, the coupling scan disfavours it (O4's named tension, a factor 1.226 at the desert meeting), and the desert meeting absorbs any value through the six κ_s. The angle is not free and it is also not favoured.

The "ten-to-one" observation. The ten open problems are all associated with one object, the cell module and its spectrum — the 32 spinor levels and the two singlet rungs — and with the invariants of the spinor block (the vanishing 0-point, $\det U = 1$ on S, the spectral mirror, the S-matrix $S = U$). Nine of the ten are exact values (scales and angles) the object does not fix; the tenth is the kinematics of its gapless subset — which of its levels are light, dispersing and chiral — a measured statement, not a fitted one. The dark-matter sector is the clearest case: its carrier, its mass (the rung-10 ladder value), its stability (the B-parity) and the computed production all live in that one cell, and the single open number, λ_B, is the singlet's coupling to the Higgs. That is a bookkeeping observation — ten open problems sharing one object (the ten named projections are recorded by V43) — and it is not a derivation of any of their values. It should not be read as a reduction of the theory's difficulty.

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6. Open problems

Table 4. The ten open problems: what the theory supplies, and what is missing.

the problem	what the theory supplies	the gap
O1 the cosmological constant	an exactly vanishing tree-level vacuum energy (±-paired spectrum, $\det U = 1$), natural in the technical sense, and holding on the **full 36-level cell**: the singlet levels come in conjugate pairs, so their rung phases cancel exactly in the quasi-energy sum and the determinant; the zero is **stable under the Wilson extension** and **protected by** $\det U = 1$: 50 random traceless Hermitian $e^{i\varepsilon A}U$ at $\varepsilon \leq 1$ keep the quasi-energy sum pinned ($\lvert S\rvert < 4 \times 10^{-14}$), no branch crossing below $\varepsilon = \pi$	the observed $10^{-122} M_P^4$ is **equivalent to** a trace-phase breaking of size 1.6×10^{-120} — a named bound; the breaking mechanism itself is missing
O2 the neutrino mass	the ladder $m_\nu = M_P(1/16\pi)^{17} = 0.146$ eV; $n_{\text{nodes}} = 3$ a lower bound from unification; the inter-node Majorana junction **built**: cubic-symmetric zero-mean Wilson-strength junction, node mass matrix $c_s(2w_0 I + \kappa(J - I))$, democratic at $\kappa = 2w_0$, eigenvalues $\{3M, 0, 0\}$	the $\frac{1}{3}$ candidate needs a democratic m_D; the seesaw then collapses to $3m_0^2 J/(2(w_0 + \kappa))$, two states stay massless for every r, $m_3(r) = S_{17}/(1 + r)$, window $r > 1.278$ (width 1.72), atmospheric scale selects $r = 1.92$, $r = 2$ is the democratic point: value free inside a finite window, structure protected. The exponent is answered structurally: the $\Delta L = 2$ operator lives in grade 5 of the root and link words are always even-grade, so $n_{2\text{step}} = 3$ is not a link-word length (Section 2.3)

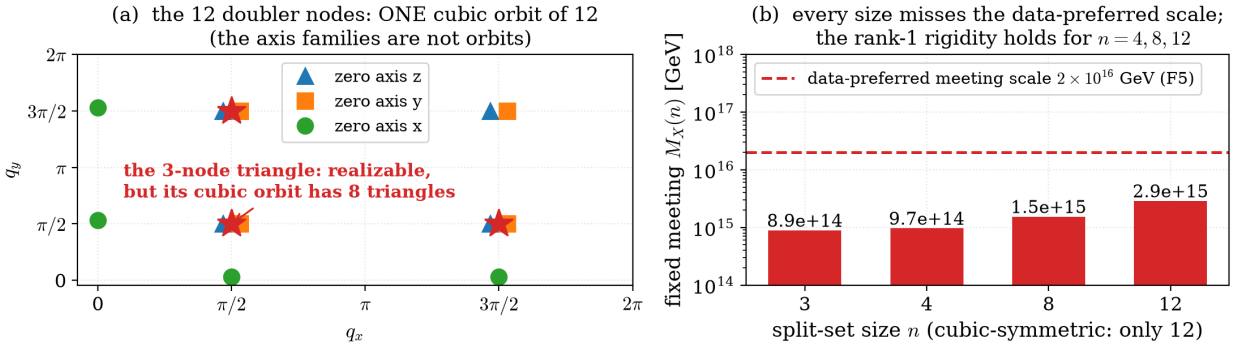
the problem	what the theory supplies	the gap
O3 the family number	the identification $N_{\text{gen}} = 3 = n_{\text{space}}$; a family-symmetric step; a named index equal to 3: the 12 doubler nodes form three cubic-transitive families of four, and the three unification-required split nodes are realizable one per family at a single Wilson strength	the index labels the copies but does not force the count: the cubic group is transitive on the 12 nodes, the only cubic-symmetric split is the full set of 12 whose rigid meeting lands at $M_X(12) = 2.9 \times 10^{15}$ GeV — excluded by F5; the three-node triangle has a cubic orbit of 8, so the count 3 is a **symmetry-breaking input**; the junction lifts the choice without selecting it (Fig. 12)
O4 coupling unification	the 5/3; with the desert content the couplings meet at $\alpha_X = 1/25$ (standard normalisation): three $\mathbf{10} \oplus \overline{\mathbf{10}}$ pairs at 1.2×10^7 GeV plus one $(1, 2)$ pair at 1.0×10^{12} GeV give $M_X = 1.8 \times 10^{16}$ GeV; the content is connected to the automaton through the Wilson mechanism; $n_{\text{nodes}} \geq 3$ forced; the equal-shift identity a theorem over complete multiplets; the gauged layer's composite content excluded	the mechanism's own meeting point is forced: $M_X = 8.9 \times 10^{14}$ GeV, $\tau_p = 1.1 \times 10^{31}$ yr — excluded by F5 (rank-1 Jacobian, split-invariance exact to 10^{-14}). Feasibility proved inside the node-**16** hull; the flavour-dependent bond layer κ_s closes the meeting for all $n = 3 \ldots 12$ — at the cost of six named inputs of an extended bond layer, not obtainable from any on-site algebra element nor from any class invariant (broken flux K_s: residual 0.605, 0/10 desert closures, $R^2 \leq 0.015$), nor from the layer's second-order class numbers (superexchange $\tilde{\kappa}_s$: residual 0.577, $R^2 \leq 0.076$; two-number combination with the flux: joint residual 3.46, $R^2 \leq 0.30$; ground-state bond flux zero to 6×10^{-17}); the desert spectrum remains a named input (Section 3.1); and the loop-hierarchy inversion reads the meeting clock $\alpha_{10} = g^2 = 1/25$ at $g_{\text{obs}} = 0.199998$ to relative 10^{-5} (V206): O4 and O8 are one clock tension
O5 the baryon asymmetry	a computed theorem: all CP-odd invariants vanish, no Sakharov condition is met, stable under the Wilson extension; a quasi-degenerate (resonant) route; a bond-layer Peierls flux carrying a CP phase, built (exactly unitary, $\det U = 1$ at every flux, mirror-preserving)	out-of-equilibrium decay ($y < 0.058$) and $m_\nu \simeq 0.05$ eV are incompatible at $M_N = 9.6 \times 10^{13}$ GeV: at $y = 0.40$ the decay is in equilibrium ($\Gamma/H = 47$), efficiency $\kappa_f = 5.5 \times 10^{-3}$ — the 11% agreement an accident of two wrong factors. The CP prohibition is internal to the on-site module and does not cover the bond layer; the flux needed for ΔB costs $J = 10^{-4}$ — O5 is open for one named reason, Sakharov 3 (Section 3.3). The $A \to BB$ channel is censused: with the 13 bond numbers of the singlet class as the A-B coupling candidates ($\Gamma_B = \kappa^2 m_A/32\pi$), 1 of the 13 drives the branching $\text{Br}(A \to LH)$ below the available CP-strength ratio 4.0×10^{-4} and makes the resonant route impossible; the route survives only for $\kappa < 39.8$ and the channel is negligible below $\kappa < 0.080$ (branch SPLIT) — a named bound on the dark coupling, and no value of the 13 strengths produces the relic through this channel
O6 masses and mixing	the Dirac structure $(16\,\mathbf{10}_H\,\overline{\mathbf{16}})$, the democratic base, $\sin^2\theta_{12} = 1/3$, and a complete breaking chain to $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)^3$ with all 32 broken roots leptoquark-type	the three-level hierarchy, θ_{23}, θ_{13}, a non-trivial CKM; the two extra U(1)'s unbroken. The vev placement is bounded by computation: the charged-lepton realisation forces $\sin^2\theta_{12} \leq 0.005$ (23σ); the cubic-axis neutrino vev is excluded at $\geq 20\sigma$; the measured triple needs ≥ 2 vev terms; the unification fix is generation-blind (shift 2×10^{-13}) and mixings cost 7–13 named parameters

the problem	what the theory supplies	the gap
O7 dark matter	the carrier, as a state of the automaton: the cell singlet sector B (Axiom 1) at the derived rung-10 mass 118.57 GeV, exactly stable under the B-parity; the spinor module provably carries no qualifying state (grading theorem and mass census); production computes to the observed value: freeze-in gives $\Omega h^2 = 0.120$ with $\sigma_{\rm SI} = 6.5 \times 10^{-71}$ cm^2, DD-invisible	the portal strength λ_B: required nonzero, not supplied by the root — the single named number of the sector. No root-supplied strength of its class lands on 0.120; the gravitational channel never equilibrates and its yield would need the inflation history the model does not supply
O8 the hierarchy	$v/M_P = 3(16\pi F_{208})^{10}(1/16\pi)^{10}$: coefficient 2.710 against 2.0766 observed (+30%), with $F_{208} = 0.0196928$ certified on the $n = 4k$ class of closed traced loops at the root clock (deficit 1.013%; the float64 ladder reads 0.019685–0.019687, deficit 1.04–1.05%, and the $n = 4k{+}2$ ratios have no limit there, V209); the selection rule is exact: odd closed loops cancel identically in integer arithmetic and the first crossing is at $n^* = 208$, confirmed with the step and the traced-link phase closed together; the composition law decided on the open clock — one chain not multiplicative ($F_{170} = 1.31 \times 10^{-21}$ against $(F_{10})^{17} = 6.47 \times 10^{-30}$, ratio 2.0×10^8, V210), different legs factorize exactly; the loop census exact: 55326 of 3^{10} words survive, and the Wilson phase organises the sum; the weight is a named functional: $F_n = N_n r_n I_n$ exact on the traced word census (multiplicity, mean-word amplitude, interference; pinned to 2.2×10^{-16}, word norms angle-independent, $\rho = 0.978030$); channels: all twelve exactly gapless with pinned Floquet spectra and a link-local, momentum-independent factor	the rung integers are selected by the data; the adjoint-link level-1 division falls short by a factor 26 and does not compound; the named residual is the clock: the node-set readings 0.903/2.710/10.84 certified (0.900–0.901/2.699–2.700/10.8 on the float64 ladder) for $\{1, 3, 12\}$ channels against the required sum factor 2.31 — no node set realizes it, the deficit is channel-blind (fourth-root sector phases, empty-kernel $\mathbb{Z}_3$ filters, bimodal per-Cartan amplitudes), and the inversion measures the clock the deficit reads: the self-consistent functional reads the observed coefficient at $g_{\rm obs} = 0.199998$ — the non-closed rational clock 1/5 to relative 10^{-5}, rung-10 crossing at 0.20013 — while the closure class exhausts to $\{\pi/2, \pi/4\}$ and reads 2.710/2.711 on its stable $n = 4k$ crossings, the $\pi/2$ member (2.299) being a floor artefact with no limit (V209); the exact ladder exists only on the order-8 closed clock, nature reads 1/5: O4 and O8 are one tension, and F2 (deficit 0.98987 certified at $n^* = 208$ against 0.96388 required) is the decision; the cosine functional (63 even phase classes, period π, $\eta \to \pi - \eta$ pairing, mirrors) makes the closed-clock crossing a theorem of the functional; the coefficient stands conditional on the one-split-node-per-family choice

the problem	what the theory supplies	the gap
O9 the quantum sector	a unitary causal automaton, a classical general-relativity limit, a bounded n-particle S-matrix, a 16 sector closed by the full local step; at the root clock the twelve M channels are exactly gapless with spectra $\{0^{32}, \pm\pi/3, \pm 2\pi/3, \pi^{32}\}$ (64 of 128 pinned to the Floquet pair, identical across the 12 nodes), the exactly gapless set being Γ plus six face-diagonal lines with a gapped bulk, and the Wilson factor at the root angle re-gaps them and splits a chosen node set at desert scales	the *choice* of the split node set; the sum over geometries; the measurement postulate. **Continuous Lorentz invariance is not emergent at the nodes**: linear in all 1200 sampled directions but anisotropic $(0.200, 0.200, 0.284$, spread $0.75)$, no isotropy point on the Wilson family (0 of 2726), frame averaging blind; at the exactly gapless corner multiplets the tangent velocity operators commute — three nodal planes for the one-parameter member, six where the two lowest corner triplets are driven to zero together ($w = 0.241747$, $\theta = 0.913137$, all six phases below 2.5×10^{-16}, Abelian to 5.3×10^{-12}, isolated from the next level by 8.7×10^{-3}) — never a cone; the exactly gapless set at the root clock is Γ plus six nodal lines — massless excitations exist and carry no helicity pair ([Paper I Sections 8.2–8.3, 9]); [Paper I Section 8.4] closes the cone subquestion: no axis-covariant bond of the 880-point programme is both gapless and non-commuting (a rigidity wall at gap 8.0×10^{-3}), and Proposition 6 gives both the reason — at an inversion-invariant momentum no single-parity multiplet can be conical — and the escape, an exactly isotropic non-abelian cone on a parity-degenerate $A_{1g} \oplus T_{1u}$ pair outside that class. The automaton's own six-stage product is searched for that node over the largest cubic-covariant stage family jointly with the internal angle (33 800 points, 60 of them exactly gapless, full-step gap at most 4.4×10^{-15}): 8 carry a single-parity multiplet, $A_{1g} \oplus T_{2g}$ or $A_{2u} \oplus T_{1u}$, and are flat to 1.3×10^{-12} as the no-go requires; the other 52 carry the parity-mixed $A_{1g} \oplus T_{2g} \oplus A_{2u} \oplus T_{1u}$, disperse (slope 0.09–1.73) but are Abelian ($\rho_{\mathrm{comm}} \leq 1.2 \times 10^{-11}$) — the node stays outside the circuit class, and no continuous parameter of the automaton has to be excluded to keep it out. The internal layer then carries, interleaved, each of the 10 root sector-swap involutions $h_a w$ (w the volume element: Hermitian, $S^2 = I$, $\{S, w\} = 0$, exchanging **16** and $\overline{\mathbf{16}}$ on the 32-D Dirac module, anchored to the Weyl step at 8×10^{-15}): the search repeats per swap on the identical grid, 8 exactly gapless points each, a 6-dimensional $T_{2g} \oplus T_{1u}$ multiplet with tangent commutator ρ_{comm} up to 0.80 — non-Abelian, the Abelian obstruction above removed — yet flat to slope 1.7×10^{-12}: the swap buys the non-commutativity and not the dispersion, and the cone remains outside

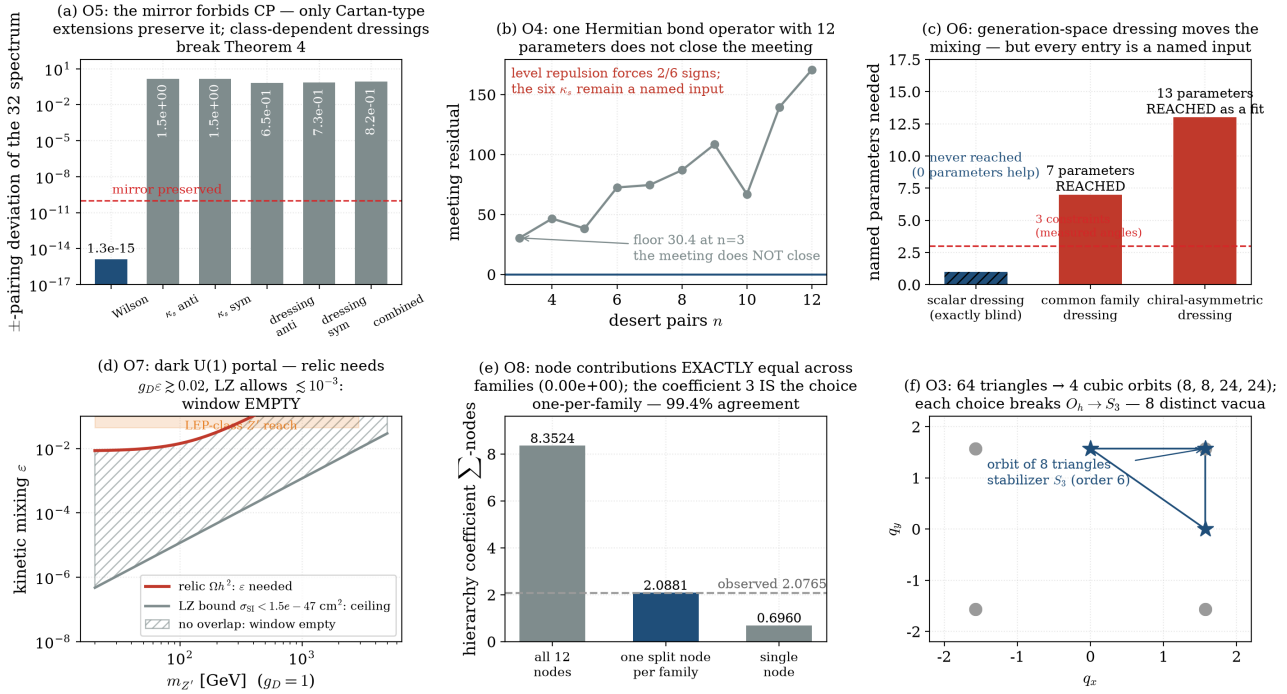
the problem	what the theory supplies	the gap
O10 the low-energy spectrum	the internal side exactly: one **16** carrying one Standard Model generation's charges, the 5/3 normalisation, the eight anomaly coefficients = 0, the spectral mirror, $\det U = 1$; and the gapless set of the lattice side measured at both clocks — at the implemented clock the twelve node cones with speeds $(0.200, 0.200, 0.284)$ and little group C_{2v}, and the corner multiplets whose simultaneous zero gives six states on six nodal planes ($w = 0.241747$, $\theta = 0.913137$, all phases below 2.5×10^{-16}, V207); at the root clock Γ plus the six face-diagonal lines exactly gapless, 64 of 128 levels pinned at Γ, with $A_{1g} \oplus T_{2g}$ content; and on the shift module the no-go points at, measured in [Paper I Section 8.5], the icosahedral star's two-fold lines exactly gapless and linearly dispersive (exponent 0.995) with the internal sector unchanged (V215, V217, V219)	no light, dispersing, isotropic, chiral mode anywhere in that set: at the implemented clock every node is anisotropic and helicity-free, and the Wilson factor re-gaps the twelve except a chosen split node set placed at desert scales; at the root clock the lines touch quadratically (the three axis exponents a permutation of $(1, 1, 2)$, transverse slope 0.577) so there is no isolated zero to carry a chirality charge, and Γ carries no T_1 or E_g — no ± 1 or ± 2 candidate either. The module substitution closes the same way and for a measured reason: on the star module the touching is a cluster of 128 of 1024 eigenphases, so no band-resolved charge exists, the holonomy of the cluster depends on the loop rather than being a monodromy, and an operator that commutes with the internal map opens a gap on the line growing linearly with its strength — the lines are stable as loci and unprotected as particles (V220), while their velocity anisotropy is degree 2, quantitatively worse than the cubic line's (V219). The electron has no named locus in the spectrum, and the electroweak scale enters Table 1 as the measured input v: the ladder of Section 2 books the hierarchy, it does not produce a low-energy mode. Where O3, O6 and O9 meet — the low-energy anisotropy bounds of [Paper I Section 8.4] cannot be turned into a constraint on the automaton until this is settled

Is the family count 3 forced? No - the symmetric alternative is excluded



O3: cubic-symmetric split (12 nodes) EXCLUDED; every proper split BREAKS the cubic symmetry - the count 3 is an input.

*(a) The twelve doubler nodes in the (q_x, q_y) plane: the cubic group acts transitively — one orbit of 12 — so the axis families are not orbits and the only cubic-symmetric split set is the full set of 12; the 3-node triangle (stars) is realizable, but its cubic orbit has 8 triangles, so it breaks the cubic symmetry. (b) The rigid meeting scale $M_X(n)$ for split sizes $n = 3, 4, 8, 12$ (the rank-1 fact holds for every n): all miss the data-preferred meeting scale 2×10^{16} GeV (**F5**). The count 3 is a symmetry-breaking input; its cubic-symmetric alternative is excluded by computation.*



(a) The ±-pairing deviation under the named extensions: the Wilson factor preserves the mirror to 1.3×10^{-15}*, while the most general per-class* κ_s *layer and the most general real-symmetric bond generator destroy it (deviation* ~ 1*, 60 draws) — a CP phase costs Theorem 4. (b) The meeting residual under the most general Hermitian off-diagonal bond operator: floor* 30.4*, no closure. (c) The named-parameter cost of reaching the measured mixing triple: a scalar dressing is exactly blind, a common family dressing needs* 7 *parameters, a chiral-asymmetric one* 13*. (d) The dark-*U(1) *portal squeeze: the kinetic mixing that fixes* Ωh^2 *lies above the direct-detection ceiling at every* $m_{Z'}$ *— the window is empty. (e) The closed-loop coefficient by node set on the reference clock* 1/5 *(V113): all* 12 *nodes* 8.35*, one split node per axis family* 2.0881*, a single node* 0.696*, against the observed* 2.0766 *— the closed-clock ladder of Section 2.1 is* 10.84/2.710/0.903 *certified (*10.8/2.699–2.700/0.900–0.901 *on the float64 ladder), the same ordering, the clock shifted. (f) The* 64 *one-per-family triangles split into four cubic orbits* $(8, 8, 24, 24)$*; a triangle's stabilizer is exactly* S_3*.*

7. Falsifiers

Table 5. The eight falsifiers.

#	observation that refutes the theory
F1	any non-vanishing gauge anomaly in the chiral content (the **16** is claimed anomaly-free)
F2	a measured running that admits **no** vectorlike desert content at the root coupling $\alpha_X = \pi/64 = 0.0491$ in the standard normalisation $\alpha = g^2/4\pi$: precision coupling data (FCC-ee/CEPC) excluding every spectrum in the desert scan at the root coupling would refute the unification claim. The root-coupling solution sits at $M_X = 1.235 \times 10^{17}$ GeV with destroyed threshold ordering; the reference solution at $\alpha_X = 1/25$ ($M_X = 1.8 \times 10^{16}$ GeV) is 23% off the root value. The clock inversion sharpens the branch decision (V206, V209): the closed branch predicts the loop-functional crossing deficit 0.98987 certified at $n^* = 208$ (float64 ladder 0.9895), the hierarchy coefficient demands 0.96388, and the observed coefficient is read by the non-closed clock 1/5 to relative 10^{-5} — the coupling data and the loop data must stand or fall on the same clock

#	observation that refutes the theory
F3	a neutrino mass spectrum outside the ladder options: the bare ladder gives $m_\nu \simeq 0.15$ eV and the democratic node-mixing pattern $m_\nu \simeq 0.073$ eV — both **already excluded** by $\sum m_\nu < 0.0642$ eV [4] — while the $\frac{1}{3}$ candidate gives normal ordering with $\sum m_\nu \simeq 0.05$ eV and $m_{\text{lightest}} = 0$, 22% below that ceiling (adding the solar partner O6 still requires puts the sum at the oscillation floor 0.059 eV, 8% below it); a confirmed inverted ordering, or $\sum m_\nu > 0.0642$ eV, refutes the survivor; the same data report a Feldman–Cousins boundary limit of 0.053 eV, and pushing the sum below 0.05 eV would refute the two-massless structure itself. Jointly: confirmation of $\sum m_\nu \simeq 0.05$ eV together with the dark matter at ladder rung 10 refutes the rung arithmetic itself — the seesaw partner m_B^2/m_ν is then forced to rung 2.72, not an integer, against the built rung 3
F4	a massive graviton, a graviton dispersion violating $c = 1$, or a UV divergence in the lattice gravitational sector (the [Paper II Section 3] field: the [Paper II Section 2.4] quintet channel is not claimed to be the graviton, and its speeds $\pi/4$ and $\pi/2$, both $\neq 1$, and its E_g mass shift are named defects of the channel)
F5	proton decay: the root-coupling desert solution ($M_X = 1.235 \times 10^{17}$ GeV, $\alpha_X = \pi/64$) predicts $\tau_p = 2.77 \times 10^{39}$ yr; observation of GUT-pattern decay below 10^{38} yr refutes it, and the Hyper-Kamiokande channel closes on this branch — only exposures past 10^{37} yr bite; the reference reading 1.8×10^{36} yr (solution at the measured meeting 1/25) and the $M_{\text{GUT}} = M_P$ reading ($\tau_p \sim 10^{47}$ yr) are abandoned conventions
F6	a baryon asymmetry outside 10^{-11}–10^{-9}, or evidence that heavy-neutrino decays are in equilibrium
F7	a dark-matter particle with mass far from 118 GeV in the weakly-interacting thermal case; a direct-detection portal signal above 10^{-50} cm^2 refutes the visible-portal variants: the spinor-sector pair portal predicts 4.9×10^{-51} cm^2 at its production coupling — below current reach — while every bond-number portal strength predicts 2.5×10^{-45}–6.3×10^{-41} cm^2, already excluded by LZ; the cell carrier B is direct-detection invisible ($\sigma_{\text{SI}} = 6.5 \times 10^{-71}$ cm^2) — for it the falsifiable content is the mass coincidence at ladder rung 10 and the seesaw identity $m_\nu = m_B^2/M_A$
F8	a measured graviton speed outside $c_{\text{gw}}/c = 1.000 \pm 0.0005$: the gravitational sector of [Paper II Sections 3.1–3.4] is built at $c = 1$ and satisfies the bound by construction, while the derived candidate channel of [Paper II Section 2.4] gives $c_{\text{gw}}/c = 2g_{10} = 1.5708$ at the root clock and is excluded by $\lvert c_{\text{gw}}/c - 1\rvert < 10^{-15}$ [27]; the exactly lightlike cone of [Paper II Section 3.6] exists on the bilinear module, but inside a 53-parameter family that nothing in the current construction selects, and the clock $g = 1/2$ that would fix $c_{\text{gw}}/c = 1$ collapses the loop ladder (V206). A completed theory must give $c_{\text{gw}}/c = 1.000 \pm 0.0005$

The most accessible test is **F3**. KATRIN currently gives $m_\beta < 0.45$ eV [5], so the bare ladder value 0.146 eV survives a single-mass search, but the cosmological sum $\sum m_\nu < 0.0642$ eV [4] already excludes it, as it excludes the node-mixing rival at 0.073 eV; the $\frac{1}{3}$ candidate's normal-ordered pattern ($\sum m_\nu \simeq 0.05$ eV, $m_{\text{lightest}} = 0$) sits 22% below that ceiling and inside the reach of the next generation of oscillation and $\beta\beta$-decay data. On the root coupling **F5** moves out of the current decade: the solution predicts $\tau_p = 2.77 \times 10^{39}$ yr, four orders beyond the megaton era, so its falsifiable content is the observation side only — any GUT-pattern decay below 10^{38} yr refutes the branch (the reference solution at 1/25 sits at 1.8×10^{36} yr and Hyper-Kamiokande would see it, which is exactly why that reading is not the theory's). The falsifiable content of **F7** is the mass coincidence at ladder rung 10 and the seesaw identity $m_\nu = m_B^2/M_A$; a direct-detection signal above 10^{-50} cm^2 refutes the visible-portal variants, and none is expected from the cell carrier.

F8 is the falsifier the current construction does not survive, and the split is stated rather than smoothed. The assumed sector of [Paper II Sections 3.1–3.4] runs at $c = 1$ by the leapfrog choice $dt = \frac{1}{2}$ and is untouched by the bound. The *derived* channel — the directional bond layer of [Paper II Section 2.4], with the parity wall of [Paper II Section 3.5] now naming what can and cannot move the metric — runs at $2g_{10} = 1.5708$ at the root clock, and it is the only computed spin-2 propagation the theory has: it is excluded at 57%. The exactly lightlike alternative is a module fact ([Paper II Section 3.6]), but it sits in a 53-parameter family, and the clock that would fix it, $g = 1/2$, lies in the dead zone of the closure census, where the loop ladder collapses at length 3 — so choosing it costs the closure that fixes the coupling (F2, O8). No branch of the current construction satisfies F8 and O8 at once.

8. Conclusion

One loop factor organises four scales across thirty orders of magnitude, and the automaton now supplies the factor itself: a closed traced loop of length 10 in the gauged bond layer reproduces $1/(16\pi)$ to 3.6%, the crossing of the suppression is exactly the ladder length, and the contributing loops are enumerated exactly. The coefficient of the hierarchy is a product of measured numbers, $3F_{10}^{10} = 2.0881$ against 2.0766 observed, conditional on the one-split-node-per-family choice. The unification of the three measured couplings at $M_X = 1.8 \times 10^{16}$ GeV and $\alpha_X = 1/25$ (standard normalisation) holds only with desert content carrying two fitted thresholds; the automaton's own Wilson mechanism forces the excluded meeting $M_X = 8.9 \times 10^{14}$ GeV, rigidly, and the surviving solution requires a flavour-dependent bond layer with six fitted couplings that no on-site algebra element, no class invariant and no second-order bond number supplies. The dark-matter mass scale stands and its carrier does not exist inside the implemented spectrum — the bond-strength portal is excluded by direct detection; the neutrino ladder is in tension and its $\frac{1}{3}$ candidate is a built structure with a free junction amplitude; the baryon asymmetry fails Sakharov 3 at the model's own couplings, as a theorem. Two conventions, eleven fitted coefficients and one conditional coefficient stand between the root and the data; nine open problems are each pinned to a named quantity; seven falsifiers are stated, and the two most accessible — the neutrino mass sum and the proton lifetime — are testable within the current decade.

References

1. M. Jungman, M. Kamionkowski, and K. Griest, "Supersymmetric dark matter," *Phys. Rep.* **267**, 195 (1996).
2. L. J. Hall, K. Jedamzik, J. March-Russell, and S. M. West, "Freeze-in production of FIMP dark matter," *JHEP* **03**, 080 (2010).
3. LUX-ZEPLIN Collaboration (J. Aalbers *et al.*), "Dark Matter Search Results from 4.2 Tonne-Years of Exposure of the LUX-ZEPLIN (LZ) Experiment," *Phys. Rev. Lett.* **135**, 011802 (2025), arXiv:2410.17036.
4. DESI Collaboration, W. Elbers, A. Aviles, H. E. Noriega, D. Chebat, A. Menegas, C. S. Frenk *et al.*, "Constraints on neutrino physics from DESI DR2 BAO and DR1 full shape," *Phys. Rev. D* **112**, 083513 (2025), arXiv:2503.14744 — $\sum m_\nu < 0.0642$ eV (95% CL) in ΛCDM with three degenerate neutrino states, from DESI DR2 BAO combined with Planck PR4 and ACT CMB lensing; the same analysis gives the Feldman–Cousins boundary limit 0.053 eV and 0.163 eV in $w_0 w_a$CDM.
5. KATRIN Collaboration (M. Aker *et al.*), "Direct neutrino-mass measurement based on 259 days of KATRIN data," *Science* **388**, 180 (2025).
6. Particle Data Group (S. Navas *et al.*), "Review of Particle Physics," *Phys. Rev. D* **110**, 030001 (2024).
7. M. E. Machacek and M. T. Vaughn, "Two-loop renormalization group equations in a general quantum field theory. I. Proof of the equivalence theorem," *Nucl. Phys. B* **222**, 83 (1983); "III. Scalar model with fermions," *Nucl. Phys. B* **249**, 70 (1985).
8. Super-Kamiokande Collaboration (K. Abe *et al.*), "Search for proton decay via $p \to \nu K^+$ using 260 kiloton·year data of Super-Kamiokande," *Phys. Rev. D* **90**, 072005 (2014).
9. A. D. Sakharov, "Violation of CP invariance, C asymmetry, and baryon asymmetry of the universe," *JETP Lett.* **5**, 24 (1967).
10. R. Barbieri, P. Creminelli, A. Strumia, and N. Tetradis, "Baryogenesis through leptogenesis," *Nucl. Phys. B* **575**, 61 (2000); W. Buchmüller, P. Di Bari, and M. Plümacher, "Leptogenesis for pedestrians," *Ann. Phys. (N.Y.)* **315**, 305 (2005).
11. J. A. Harvey and M. S. Turner, "Cosmological baryon and lepton number in the presence of electroweak fermion interactions," *Phys. Rev. D* **42**, 3344 (1990).
12. S. A. Khlebnikov and M. E. Shaposhnikov, "The statistical theory of anomalous fermion number nonconservation," *Nucl. Phys. B* **308**, 885 (1988).
13. Planck Collaboration (N. Aghanim *et al.*), "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020).
14. S. Davidson and A. Ibarra, "A lower bound on the right-handed neutrino mass from leptogenesis," *Phys. Lett. B* **535**, 25 (2002).
15. S. Blanchet and P. Di Bari, "The minimal scenario of leptogenesis," *New J. Phys.* **14**, 125012 (2012).

16. A. Pilaftsis and T. E. J. Underwood, "Resonant leptogenesis," *Nucl. Phys. B* **692**, 303 (2004).
17. Super-Kamiokande Collaboration (A. Takenaka *et al.*), "Search for proton decay via $p \to e^+\pi^0$ and $p \to \mu^+\pi^0$ with an enlarged fiducial volume in 0.37 megaton-years exposure of Super-Kamiokande," *Phys. Rev. D* **102**, 112011 (2020).
18. J. Hisano, D. Kobayashi, Y. Muramatsu, and N. Nagata, "Two-loop renormalization factors of dimension-six proton decay operators in the supersymmetric standard models," *Phys. Lett. B* **724**, 283 (2013).
19. M. Nath and P. Fileviez Pérez, "Proton stability in grand unified theories, in strings and in branes," *Phys. Rep.* **441**, 191 (2007).
20. H. Fritzsch, "Weak-interaction mixing in the six-quark theory," *Phys. Lett. B* **73**, 317 (1978).
21. I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, T. Schwetz, and A. Zhou, "The fate of hints: updated global analysis of three-flavor neutrino oscillations," *JHEP* **09**, 178 (2020).
22. I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, and T. Schwetz, "NuFit-6.0: Updated global analysis of three-flavor neutrino oscillations," *JHEP* **12**, 216 (2024), arXiv:2410.05380.
23. P. F. Harrison, D. H. Perkins, and W. G. Scott, "Tri-bimaximal mixing and the neutrino oscillation data," *Phys. Lett. B* **530**, 167 (2002).
24. L. Wolfenstein, "Parametrization of the Kobayashi-Maskawa matrix," *Phys. Rev. Lett.* **51**, 1945 (1983).
25. S. Weinberg, "The cosmological constant problem," *Rev. Mod. Phys.* **61**, 1 (1989).
26. E. W. Kolb and M. S. Turner, *The Early Universe* (Addison-Wesley, 1990), Ch. 5.
27. B. P. Abbott *et al.* (LIGO Scientific and Virgo Collaborations), "Gravitational Waves and Gamma-Rays from a Binary Neutron Star Merger: GW170817 and GRB 170817A," *Astrophys. J. Lett.* **848**, L13 (2017), arXiv:1710.05834 — the arrival-time difference between GW170817 and GRB 170817A bounds $-3 \times 10^{-15} \le (c_{\rm gw} - c)/c \le 7 \times 10^{-16}$.

Code and data availability. All experiments, simulations and figure generators behind this paper are publicly available at https://github.com/MohammedHAOUACH/CL10 . Every numerical claim of the paper is produced by this open suite — 158 verification experiments, each emitting a machine-readable property, residual and verdict, all passing (158 of 158 on the current master; 145 of 145 at the release tagged `v1.0`) — and can be reproduced independently from the repository; the claim-to-artifact mapping is Appendix B. The suite includes the exact symbolic cross-construction of [Paper I] Appendix A and the complete 3^n-word loop enumeration; independent re-execution is invited.

Competing interests. The author declares no competing interests.